

Finality Assurance™ for Reliable Decision Quality

Runtime Conflation, Governed Routing, and Computed Reliance

The core technical article: the problem, the formal result, the architecture — Finality Assurance™ (FAS), Governed Multi-Route Selection (GMRS), Constrained Policy Reinforcement Learning, and the Mechanistic Conflation Engine (MCE) — the benchmark evidence, the economics of Accepted Work, and the governed work-management product, in one sitting.

What may a named party rely upon, about a named result, for a named purpose, now — computed, not written.

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Qualified against the monograph release candidate identified by digest in reference [1]; the release digests of both subjects are recorded in the release custody record. Every technical claim, number, boundary, and reservation is carried from that monograph (reference [1]) without strengthening, and this article is self-contained. Nothing here is a commitment, availability claim, or field-validation result.

Abstract

What changed. Artificial intelligence made candidate output abundant, not defensible reliance. Organizations still decide whether consequential work can be used through mutable status fields, prose, self-attestation, fragmented tools, and bounded human review — mechanisms that neither scale nor keep distinct questions distinct. The governing distinction is **generation** (a candidate) versus **evaluation** (a score) versus **acceptance** (a named party may rely on it, for a named purpose, now) — the third is the scarce act. **The structural defect** is representational, not cultural: a status field is memory, not present measurement. Over the elected twelve-state model, any status vocabulary of fewer than six values necessarily collapses permission against effect status, or effect status against reliance license — and with an unconstrained write path, enlarging the vocabulary repairs nothing: the defect is in the write path.

The architectural answer is to compute the answer instead of writing it: Finality Assurance™ represents obligations, evidence, authority, occurrence, currentness, assertion, and reliance as separate typed objects; derives a categorical verdict by conservative composition over an anchored roster; licenses claims and reliance separately; and recomputes when any basis changes. GMRS governs execution across routes; constrained policy learning adapts without defining its own admissibility; MCE converts discovered failure patterns into machine-enforced controls. **What this contributes to computer science.** Five bounded contributions (section 3.6): conflation as a property of representation and settability; proof obligations for decision composition; observation assurance distinct from consensus; measurable assurance predicates; explicit proof, observability, decidability, and trust boundaries. These five characterize the article's computer-science contribution; the later sections instantiate, operationalize, or test the work from which they are stated.

What has been measured. Across 2,520 graded answers from six real language models on verifiable tasks, the ungoverned accept-on-claim arm shipped seven wrong or fabricated answers; the governed arm shipped none. On a 30-task legitimate-work battery: 0 of 270 correct, evidenced answers refused, 90 of 90 correct unevidenced answers refused — a designed friction cost published with the benefit. On a recorded, saturated 36-task battery that cannot resolve a quality difference: the strongest static comparator matched with 24 rather than 36 model calls — twelve exact deterministic substitutions, no frontier calls. On a paired 240-task live campaign, most modeled savings arose where outcomes were identical — 75 identical correct refusals at under half the frontier comparator's cost, declared list-price model. **What the adverse live result showed.** On that same campaign the governed route recorded 21 false accepts against the comparator's 1, and 0.883 correct disposition against 0.988. End-to-end superiority is not established by this live campaign — it is the current operating envelope and the follow-on study's target. All results are producer-graded on named local estates; no third party has independently validated anything. **What remains to be established.** Bounded benchmark evidence and sustained internal operation support benefits from the governed adaptation architecture and the surrounding routing and control system within the estates reported here. The public evidence does not isolate the incremental contribution of adaptation itself over time or establish its generalization to independently authored external estates; those are targets of the powered benchmark and independent external validation. Implementation-specific promotion history and update-control evidence are reserved.

Potentially life-saving and mission-critical applications, and the research still required. It may matter most where stale, unsupported, or unauthorized reliance can contribute to patient harm, unsafe physical control, infrastructure interruption, or unauthorized mission action: treatment authorization and medical-device release, air-traffic and rail control, nuclear operations and grid protection, defense mission systems. No deployment, efficacy, safety, certification, or outcome is reported in any of them, and computed reliance

replaces none of the hazard analysis, integrity-level verification, timing determinism, barrier independence, regulatory approval, clinical judgment, or operational command those domains require; it must earn its place through the validation agenda of section 9.1 — an invitation to that research, not a claim it has been done.

Principal results

Every result travels with its boundary; the boundary is part of the row, and no cell is quotable without the others in its row. All rows are producer-graded on named local estates; no third party has independently validated anything; no field validation exists.

Table 1. Principal results. Adverse rows carry the same prominence as favorable rows.

Finding	Why it matters	Evidence	Boundary
Decision quality. Computed acceptance shipped zero wrong or fabricated answers; accept-on-claim shipped seven.	The mechanism changes which outputs become reliable work at the point of acceptance.	2,520 graded answers, six real language models, paired task-for-task; 7 vs 0 false accepts.	Recomputable, verifiable tasks; a soundness property of recomputation; not a general accuracy claim; no field result.
Evidence-qualified admission. Evidence presence, rather than answer correctness alone, determined admissibility.	Required evidence functions as a real admission condition, not decoration.	0 of 270 correct, evidenced answers refused; 90 of 90 correct-but-unevidenced answers refused.	Separate 30-task legitimate-work battery; the 90/90 refusals are the designed, published cost of the discipline.
Recorded route composition. Governed routing matched the strongest static comparator with one-third fewer model calls.	Some valid dispositions are reachable through cheaper, exact routes.	36-task frozen battery; 24 vs 36 model calls; 12 exact deterministic substitutions; 0 frontier calls; 1,000 correct disposition both compared arms.	The battery saturated: it cannot resolve a quality difference and licenses no parity claim.
Live outcome-equivalent economics. Most modeled savings occurred where the outcome was identical.	Most of the modeled savings did not depend on accepting a worse outcome.	Paired 240-task live campaign; 75 identical correct refusals at less than half frontier cost; ~96.5% of modeled saving on same-outcome pairs.	Declared list-price cost model, which diverges materially from observed usage cost; descriptive, not powered confirmatory evidence.
ADVERSE — current live operating envelope. The governed route lost quality and safety on the live campaign.	A routing layer that could not surface this would be worse, not better; the boundary is the finding.	21 false accepts against 1; 0.883 correct disposition against 0.988, same 240 tasks.	End-to-end superiority is not established by this campaign; this is the target of the follow-on study.
Governed adaptation. Bounded internal and benchmark evidence supports the governed adaptation architecture; its incremental longitudinal contribution is not yet isolated.	Adaptation is being evaluated inside constraints it cannot define for itself, while the surrounding governed system has already produced measurable decision-quality, evidence-discipline, routing, economic, and operational benefits.	Bounded benchmark, prototype, and sustained internal-operational evidence described in this article and the monograph; implementation-specific promotion history is reserved.	The magnitude, persistence, and causal increment attributable specifically to adaptation, and its generalization to independently authored external estates, remain to be established.
Supporting assurance.	The controls that produce the numbers were themselves adversarially exercised.	360/360 planted hazards blocked (matched clean arm); 46/46 release-regression mutants detected; 8 paired mutants over 2 shared clean-control executions; 53/53 and 109/109 declared outputs regenerated.	Causal attribution follows for the 8 paired mutants only; 4 of 6 hazard scenarios share one window-opening mechanism; author-side regeneration is not independent validation.

1. The question, carried through one example

A team wants to deploy commit X. **Admission** asks whether deployment may begin under current authority and policy. **Execution** asks whether the deployment process ran. **Occurrence** asks whether commit X is actually present on the intended production hosts — a different fact, independently observed. **Outcome** asks whether the intended condition followed: the service is healthy and no prohibited effect occurred. **Canonicity** asks which record controls when the build system, the deployment platform, the ticket, and the monitoring system disagree. **Assertion** asks what the evidence permits anyone to say — deployed to ten percent of hosts is not release complete. **Reliance** asks whether a named release manager may route customer traffic on the strength of the record, now. **Applicability** asks whether yesterday's answer still binds after the world has moved.

Every one of those questions is routinely answered today by a single settable field, a green check, or a sentence in a ticket. The same eight questions attach unchanged to a wire transfer, a batch release, or a prior authorization; commit X is the worked instance, not the category. The question this article answers is the one that a regulator, an auditor, or a counterparty will actually ask:

What may a named party rely upon, about a named result, for a named purpose, now?

1.1 The vocabulary this article needs

Seven objects are enough to read the rest of the paper; each is developed at its point of use. The **obligation roster** is the anchored, versioned list of what must be true before anyone may rely — declared in advance by an authority separate from the work, and not shrinkable by the caller. The **Finality Account** is the complete retained case file for one governed decision: every dimension, blocker, and piece of supporting proof kept separate rather than averaged into a score. The **Finality Verdict** is the categorical answer computed from the account for one exact reliance question; it is not settable by anyone. A **claim license** bounds what may be said about a result; a **reliance license** bounds who may act on it, for what, and until when — different objects, deliberately. **Accepted Work** is work actually permitted to count under declared acceptance conditions. **Cost-to-disposition** is everything it took to reach any disposition at all.

Two things this article is not. It is not a summary of the monograph: it is the monograph's causal and evidentiary spine, restated whole at a scale a reader can hold — problem, formal defect, architectural escape, measured evidence, economics, applicability, product realization, open questions. And it is not a product announcement: every capability statement is typed as a design commitment, every measurement is bounded to its named campaign, and the boundaries travel with the claims wherever they are quoted.

A note on reading the evidence sections. Results in this article are stated with their full strength and their full boundary in the same breath — including one deliberately prominent adverse result. That is not modesty; it is the article's own thesis applied to itself. A reliance layer that asks to be trusted on the strength of its own unbounded claims would have refuted itself.

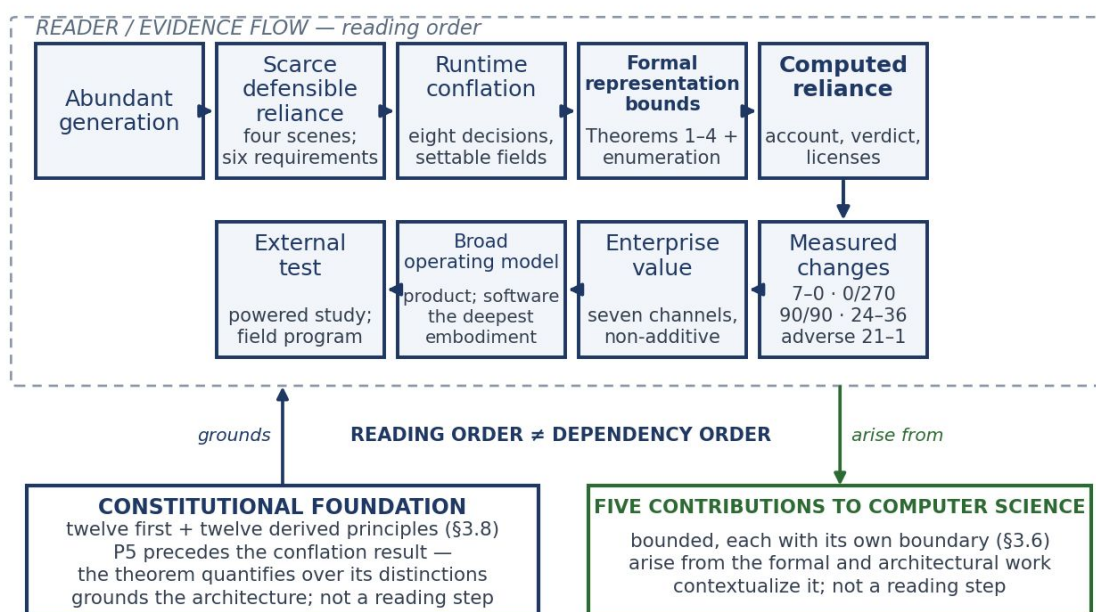


Figure 1. The reader's path, and what it rests on. The framed band is reading order; the two boxes below are dependency, not sequence — the principles ground the architecture (P5 precedes the conflation result, supplying the distinctions the theorem quantifies over), and the five contributions arise from the formal and architectural work. Reading order is not dependency order.

1.2 One hierarchy, three zoom levels

The article uses a single hierarchy throughout. Five reader questions are the plain-language account; eight runtime decisions are those questions resolved to the granularity a runtime needs; three theorem coordinates are the minimal formal model in which the representation result is proved. These are zoom levels of one object, not competing taxonomies.

The compression is deliberate at each level. The five questions are what a reader must be able to ask without training: allowed? happened? intended result? what may be said? who may act on it? The eight decisions are what a runtime must keep separate to answer those questions without merging them. The three coordinates are the smallest algebra in which the merging itself can be proved unavoidable for small vocabularies. Nothing at a finer zoom contradicts a coarser one; a reader who retains only the five questions and the one boxed question above can follow every section that remains.

Table 2. The reader's crosswalk: the five questions, the eight runtime decisions, and the three theorem coordinates. The theorem's coordinates are a deliberately narrow core, not a competing inventory of the eight.

Reader-facing question	Runtime decision(s)	Theorem coordinate
Was this allowed?	Admission	Admission (A)
Did it happen?	Execution, and Occurrence	Composite effect status (O)
Did the intended result follow?	Outcome support	not separately modeled
What may be said?	Canonicity, and Assertion	not separately modeled
May this party act now?	Reliance, and Applicability	Reliance (R), with currentness handled separately

The minimal model deliberately compresses: it keeps exactly the coordinates needed to prove that common representations must collapse distinctions, and it treats each decision as correctly attributed, timed, and scoped — assumptions the formal section then removes one at a time.

2. Why the operating model must change

Four scenes, one structure. A physician's practice spends thirteen hours of physician and staff time every week — the American Medical Association's 2025 physician survey, published May 2026 [2] — assembling prior-authorization proof. The survey establishes the time and the request volume; that a material share of those hours is evidence reconstruction — regathering facts that already exist somewhere — is this article's hypothesis about their composition, not the survey's finding, and field work should measure the actual allocation. A trading desk watches an opportunity expire because the confirmation that would make acting defensible takes longer than the window in which acting is possible. A mission operator holds an authorization that was valid when issued — and has no mechanical way to establish that it is still valid now. A production line's release depends on a sensor whose calibration lapsed; the batch record says released, and nothing in the record knows about the sensor. Four industries, one structure: something was done, somebody must rely on it, and the record of what happened cannot carry the weight being put on it.

Look closer at what the four scenes share. In each, the work itself is not the problem — the authorization was issued, the analysis was done, the batch was produced. The problem is that the record standing between the work and the party who must act on it answers a different question than the one being asked. The prior-authorization binder answers what was assembled, not whether the payer's current policy is satisfied now. The confirmation chain answers who signed, not whether the basis still holds inside the trading window. The mission authorization answers what was approved then, not what is valid now. The batch record answers what was logged, not what the lapsed sensor can no longer support. Each estate compensates with people — and the compensation is invisible precisely because it works, until volume makes it fail.

The dilemma that follows is the delegation dilemma, and it is the economic engine of this article. An organization facing machine-speed production has two ways to keep decisions defensible: refuse to delegate consequential work to machines, forfeiting the production abundance; or accept results on the producer's claim, importing silent risk at machine speed. Every scene above is an organization paying to avoid choosing — with reviewers, reconciliation, and reconstruction. The claim of this article is that the choice is false: the missing instrument is an acceptance decision that is computed, evidence-derived, and recomputed as the world moves.

Take the healthcare scene at working depth. The thirteen hours are not spent healing anyone; on the stated hypothesis about their composition, a material share is spent reconstructing facts that already exist — the indication in the chart, the policy in the payer's manual, the license in the registry — into a form a distant reviewer will accept. Every element of that binder is evidence gathering against an obligation roster nobody has written down, graded by an authority whose criteria arrive as denial codes. The work is real; the waste is that it is performed by hand, per-request, after the fact, and thrown away. The same description fits the trading desk's confirmation chain and the operator's revalidation checklist almost word for word — only the domain contract changes.

The manufacturing scene adds the currentness lesson. Nothing was wrong with the batch record when it was written; the sensor's calibration lapsed afterward, and the record kept asserting what it could no longer support. Paper records cannot be expected to know this. Computed records can: if reliance on the batch depends on the calibration, the dependency is recordable, the lapse is detectable, and the release decision can be recomputed — before shipment rather than in the recall.

2.1 Generation abundance, reliance scarcity

Model accuracy and specific-result reliance are different questions. An executor that is right 95 percent of the time tells you nothing about whether **this** result, produced at machine speed among thousands, is one a named party may act on. As generation becomes cheap, the binding constraint moves to the acceptance decision — and the acceptance decision is still made with instruments built for human-paced work: status fields, sign-offs, dashboards, and prose. Human assurance capacity is bounded: expert review does not scale with machine production, so organizations either decline to delegate consequential work or accept results on the producer's claim. Both choices are expensive, and the second is silently dangerous.

2.2 The telephone game

Underneath every fragmented estate the same process runs continuously: a message whispered along a line. Work is done under one system's vocabulary; its result is compressed into a status; the status is copied into another tool's vocabulary; a dashboard summarizes the copies; a person summarizes the dashboard. Each hop compresses what was known into what can be repeated, nobody in the chain can tell that it happened, and the party who finally acts is relying on the last retelling rather than on the original. Synchronizing status fields across systems does not repair this; it propagates it.

The telephone game also explains why integration projects disappoint: wiring the tools together copies statuses faster without making any of them mean more. When the build system, the deployment platform, the ticket, and the monitoring system disagree, synchronization guarantees only that they will disagree in unison later. What is missing is canonicity — a declared answer to which record controls — and no amount of copying manufactures it. That is why the architecture in section 4 begins from declared obligations and typed evidence rather than from connectors.

2.3 Trust reconstruction is the de facto trust layer

What organizations do today is not nothing — it is competent compensation. Reruns, reconciliation meetings, approval chains, audit reconstruction, retries, second reviews, and rework are the actual trust layer of modern operations: expensive, slow, and applied after the fact precisely because the records produced during the work cannot answer the reliance question on their own. That compensation spend is real money and real expert time, and it grows with production volume — which is exactly what machine production makes explode.

The capacity arithmetic is unforgiving. Expert review is the scarce input: it does not scale with production, and it is exactly the resource consequential domains are shortest of. When production multiplies and review does not, one of three things happens — review becomes sampling, sampling becomes attestation, or the queue becomes the product. All three convert the acceptance decision into an assumption. An acceptance decision that is computed changes the slope: expert attention is spent where the evidence fails or the stakes demand judgment, not uniformly across everything produced.

2.4 What any adequate solution must answer

Whatever the architecture, a solution to this problem must answer six questions. This list is the standard; it is worth arguing with on its own terms.

1. What was required before this decision could be relied upon — declared in advance, by an authority separate from the work?

2. What discharged each requirement, in a form that survives inspection and is separable from whoever produced the work?
3. Did it actually occur — established through a path capable of contradicting the executor's own account?
4. Who may rely on it, for what, and until when — rather than a single global claim addressed to nobody?
5. Is it still current now — and how is reliance unwound, with notice, when its basis fails?
6. What did reaching the disposition cost — fully loaded, not just the compute line?

Ordinary status fields, prose, and self-attestation do not by themselves answer all six requirements. The next section formalizes one central representation failure for settable status projections; later sections address the remaining evidence, authority, currentness, and reliance obligations. Nothing later in this article depends on accepting the scenes above; the formal result stands on its own.

The six requirements also give the reader a bounded diagnostic exercise. Take one decision your organization makes routinely and would struggle to defend afterwards — a release to production, a figure in a filed statement, an approval, a batch release. Ask whether the obligations for that decision can be written down; whether the evidence classes that discharge them can be named; whether that list would stay stable long enough to be worth maintaining. Where all three answers are provisionally yes, the decision is a candidate for governed finality — applicability still depends on roster completeness, evidence availability, authority, conflict semantics, and maintainability — and many organizations may find that relevant obligations are already distributed across policies, standards, contracts, and runbooks. Whether those sources form a complete and maintainable roster for a named decision is a field question; the engineering opportunity is to compute against such an anchored roster at reliance time rather than reconstruct it afterward.

Preview the shape of the answer against those six requirements, because the architecture section will discharge them in order. What was required becomes the anchored obligation roster. What discharged it becomes typed evidence with declared ceilings. Did it occur becomes an occurrence path capable of contradicting the executor. Who may rely, for what, until when becomes the reliance license — named party, named purpose, bounded window. Is it still true becomes Continuous Finality™ — dependency-driven recomputation with withdrawal and notice. What did it cost becomes cost-to-disposition, accounted against Accepted Work. None of the six is satisfied by a better status field, and all six are satisfiable by computation over declared objects — that is the article's engineering claim, and the sections that follow are its discharge.

One more framing prevents a common mispricing. The costs of the status quo are rarely booked as what they are: reconciliation meetings, audit preparation, evidence re-assembly, redundant review, and post-incident reconstruction are distributed across cost centers and normalized as how work is done. Summed, they are a reliance tax — paid continuously, at expert rates, for a capability the estate still does not end up having, since none of that spend makes the next decision defensible by construction. The economic comparison in section 6 is therefore not new spend against zero; it is computed reliance against a tax already being paid in the least visible currency the organization has: its experts' attention.

3. Runtime conflation, and what is provable about it

3.1 Status is memory, not present measurement

The usual model for recording that work is finished is a row with a flag on it: created, read, updated, deleted, and somewhere in the columns a boolean or a short enumeration meaning accepted, approved, done. That flag records that some process once reached a conclusion. It does not check whether the conclusion still holds. Four defects recur, and they are independent of one another: the status can be **stale**, because the conditions that justified it have changed while the record remained byte-identical; it can **conceal uncertainty**, because a categorical label absorbs a distribution; it can be **self-authored**, because the party with the strongest incentive to declare success is frequently the party permitted to write the field; and it can **persist after correction**, because correcting the underlying evidence does not automatically propagate to every place the conclusion was recorded or relied upon.

3.2 Eight runtime decisions, varying independently

Admission, execution, occurrence, outcome support, canonicity, assertion, reliance, and applicability are not ordinary sequential stages of one pipeline. Each can vary independently of the others: a run can execute without its effect occurring; an effect can occur without authorization; an authorized, occurred, verified effect can still carry no license for a particular party's reliance; and any of these can be current or stale independently. **Runtime conflation** is the collapse of two or more of these decisions into one recorded dimension, so that the record can no longer distinguish situations the organization is obligated to distinguish. Plainly: one field is asked to hold several different answers, and it can only hold one.

Table 3. The eight runtime decisions. Not sequential stages: each varies independently.

Decision	The question it answers	Typical conflation today
Admission	may this work begin, under current authority and policy	merged into assignment or ticket state
Execution	did the process run	treated as proof the effect occurred
Occurrence	did the intended effect actually happen, independently observed	inferred from the executor's own report
Outcome support	did the intended condition follow, with no prohibited effect	collapsed into a green status
Canonicity	which record controls when systems disagree	undeclared; loudest dashboard wins
Assertion	what may be said about the result	press-release grammar over partial facts
Reliance	who may act on it, for what, until when	one global DONE addressed to nobody
Applicability	does yesterday's answer still bind today	staleness discovered at incident time

3.3 The minimal formal model

Elect three coordinates, each with its smallest useful vocabulary:

$$\begin{aligned}
 A &= \{\text{REFUSED}, \text{GRANTED}\} \\
 O &= \{\text{NOT_OCCURRED}, \text{OCCURRED_VERIFIED}, \text{UNAUTHORIZED_EFFECT}\} \\
 R &= \{\text{UNLICENSED}, \text{LICENSED}\}
 \end{aligned}$$

$$G = A \times O \times R \quad |G| = 12$$

G is the full product: all twelve governance states are treated as representable. Where a deployment can argue that some states are unreachable, the theorems restate over the admissible subset G_{valid} ; the bounds below then read with $|G_{\text{valid}}|$ in place of $|G|$. This is an elected model, not an unavoidable ontology: the claim defended here is that these coordinates are distinctions consequential work is obligated to keep, not that no other model exists. A **status vocabulary** S with projection $\pi: G \rightarrow S$ represents these states; the question is what any small S must destroy.

Two collision classes are named, for distinct states g, h sharing a projection (a fiber of π):

$$\begin{array}{ll} C1(g, h) \Leftrightarrow A(g) = A(h) \wedge O(g) \neq O(h) & \text{— permission and effect status merged} \\ C2(g, h) \Leftrightarrow O(g) = O(h) \wedge R(g) \neq R(h) & \text{— effect status and reliance license merged} \end{array}$$

Safety is defined relative to a predicate set and carries no meaning apart from one: a fiber is safe under $\{C1, C2\}$ if no pair inside it instantiates either collision.

A concrete pair makes the classes vivid. Let g be (GRANTED, OCCURRED_VERIFIED, LICENSED) — authorized work, verified effect, reliance licensed — and h be (GRANTED, UNAUTHORIZED_EFFECT, LICENSED): same permission, same license, but the effect that occurred was never authorized. A vocabulary that assigns g and h one label has committed $C1$ — permission merged with effect status — and the estate cannot see the difference between a deployment that went as approved and one that escaped its approval. $C2$ is the subtler merge: two states with identical effect status, one licensed for reliance and one not, sharing a label — the difference between done and safe to act on, erased.

Why these three coordinates, and not others? Because they are the minimal axes of the two collisions that external questioners actually ask about. A regulator's first two questions after an incident are was this authorized and did the stated effect actually occur — the A and O axes; a counterparty's question is was anyone licensed to act on it — the R axis. Admission, occurrence, and reliance are the coordinates on which accountability turns, which is why collapsing them is the expensive merge. The fuller eight-decision model refines these axes (execution versus occurrence; outcome support; canonicity; assertion; applicability) without changing the counting argument, and every refinement only raises the required representational size.

A note on the fiber language, for readers new to it: a fiber of π is simply the set of governance states that share one label — everything the vocabulary treats as the same. The theorems are statements about what must end up in the same fiber. The engineering translation is direct: a fiber is a status value's blind spot, and the collision classes name the blind spots an estate cannot afford. Safety of a fiber is always relative to a declared predicate set — there is no absolute safe vocabulary, only a vocabulary safe for the distinctions an estate has declared it must keep.

Theorem 1 (forced sharing at four values). For any four-valued vocabulary, π cannot be injective: at least $\lceil 12/4 \rceil = 3$ states share some status, at least $|G| - |S| = 8$ states are excess beyond one per status, and at least 9 states sit in shared fibers. **Plain language:** with twelve situations and four labels, most situations share a label with situations they must be distinguished from. **What it does not mean:** it does not yet say which distinctions are lost — that is Theorem 2.

Formally: for $|S| = 4$ and $|G| = 12$, $\pi: G \rightarrow S$ forces $\text{max-fiber} \geq \lceil 12/4 \rceil = 3$, excess states beyond one-per-label $\geq 12 - 4 = 8$, and at least 9 states in shared fibers (the minimum is attained at the fiber profile 9-1-1-1). **What it means:** sharing is not an implementation accident to be engineered away; it is forced by counting. **What it**

does not mean: it does not yet identify which merges are harmful — a vocabulary could in principle spend its collisions on harmless pairs, which is exactly the question the collision classes and Theorem 2 make precise.

Reading the theorems correctly matters as much as proving them. Formal statement, plain reading, and non-reading are given together for the results most at risk of over-reading, because a theorem about representation must not be mistaken for a product effectiveness result. These are statements about what any status vocabulary of a given size must destroy, under an elected model whose coordinates are argued for — not measurements of any system, and not guarantees about any implementation.

Theorem 2 (lower bound under C1, C2). If every fiber of π must be safe under $\{C1, C2\}$, then $|S| \geq \lceil |G| / |A| \rceil$, and for the full product $|S| \geq |O| \times |R| = 6$. A six-valued vocabulary attains the bound — by projecting onto (O, R) and discarding admission entirely. **The bound-attaining projection is therefore not a design recommendation:** it avoids the two named collisions by refusing to represent permission at all.

What Theorem 2 means: six is the floor for keeping occurrence and reliance distinctions safe, and the floor is reachable only by paying with the admission coordinate. **What it does not mean:** it does not say six values are adequate — the bound-attaining projection is the proof of the opposite, an existence result about the price of smallness. Any real estate that cares whether work was authorized needs C3, and C3 changes the answer to twelve.

The model's remaining generosity should be stated. Everything so far assumes each decision arrives correctly attributed, correctly timed, and correctly scoped — that the writer of a status is who the record says, that the timestamp means what it appears to mean, that the label refers to the version it sits beside. The settability results remove the first assumption; currentness (section 4.5) removes the second; anchoring (section 4.1) removes the third. The formal core is thus the floor of the problem, not its extent.

Theorem 2' (adding admission preservation). Add C3: states differing in admission may not share a fiber. Under $\{C1, C2, C3\}$, every safe fiber is a singleton — $|\text{safe fiber}| = 1$ — so $|S| \geq |G| = 12$ and π must be injective. **What it means:** preserving all three distinctions categorically requires representing all twelve states; every smaller vocabulary is a decision about which distinction to destroy.

What Theorem 2' means: an estate that wants admission, occurrence, and reliance kept categorically distinct cannot buy the result with a cleverer six- or eight-value status; it needs the full twelve — or, equivalently, a representation that stores the coordinates separately, which is what a multidimensional account is. **What it does not mean:** it does not require twelve values in a user interface; projections for display are harmless exactly when the retained object underneath is not a projection.

Exhaustive corroboration. Over all $4^{12} = 16,777,216$ mappings from G into a four-valued status vocabulary, exhaustive enumeration returns zero that avoid both named collision classes. This is implementation corroboration — a machine check that the implemented predicates match the stated ones — not an independent proof of the analytic theorem, and it is reported as exactly that.

Why report the enumeration at all? Because it checks a different thing than the proof does. The analytic argument establishes the bound over the stated model; the exhaustive run establishes that the implemented predicates — the code meanings of C1 and C2 that the architecture actually enforces — match the stated ones over every one of the 16,777,216 mappings. A specification error in the predicates would have surfaced here as a nonzero count. Machine corroboration of this kind recurs throughout the architecture: the theorem carries the necessity; the enumeration carries the conformance.

3.4 Settability: the deeper defect

Theorem 3 (write-path defect). Take the exact hypothesis seriously: a status field with **unconstrained, truth-independent writability** — any enabled actor or process can write any permitted value regardless of the underlying facts. Under that hypothesis the reachable (state, label) space is the full $G \times S$, and enlarging the vocabulary restores no guaranteed relation between record and facts: vocabulary expansion cannot cure a write-path defect. **What it does not say:** it does not condemn every writable field; a field whose writes are constrained to evidence-derived transitions is outside the hypothesis. Do not read settable more broadly than the theorem states it.

What Theorem 3 means in practice: the write path, not the vocabulary, is the deeper defect. An estate that responds to conflation by enriching its status taxonomy — more states, finer workflow, better naming conventions — while leaving the field writable by any enabled actor has spent its effort on the coordinate the theorem says cannot help. **What it does not mean:** it does not forbid caches, derived views, or convenience fields; those are outside the hypothesis the moment their writes are derived from evidence rather than free.

Theorem 4 (reliance must narrow within an epoch). Sound reliance management requires $R(e, k+1) \subseteq R(e, k)$: within a reliance epoch e , corrections may only narrow the licensed set; a withdrawal must not be silently undone by a rewrite. Reliance legitimately widens sometimes — on new evidence, a new authority grant, a successful appeal, completed remediation, or a new policy epoch — so the property is indexed rather than absolute: widening requires opening a new epoch with its own identity, never a quiet rewrite inside the old one. The impossibility half rests on four assumptions the source states rather than smuggles in: the status values are interpreted through a mapping to licensed (party, decision) sets, with at least one writable value licensing the pair in question; at least one value or event withdraws it; the licensing value remains arbitrarily writable after the withdrawal; and no immutable epoch marker or revocation memory constrains the rewrite. Under those assumptions a single reachable counterexample suffices: a status sequence that revokes and then resets — CLOSED, APPROVED, CLOSED — silently re-licenses a withdrawn (party, decision) pair — undetectable where the field is the only record — so within-epoch narrowing is unenforceable by that record alone. The safe construction is the contrast: under evidence-derived reliance, every within-epoch transition is a subset step by construction — $\text{Succ}(L) = \{L' : L' \subseteq L\}$, of which there are $2^{|L|}$; over a four-element (party, decision) space all $2^4 = 16$ licensed sets are reachable from the full set, each only from a superset of itself, so widening within an epoch is structurally unreachable. Improved evidence may still raise a verdict across recomputation; the monotonicity is subset-narrowing of licensed reliance within an epoch, not a prohibition on ever improving.

Theorem 4, read the same four ways. Formally: $R(e, k+1) \subseteq R(e, k)$ within an epoch, unenforceable by a sole arbitrarily rewritable licensing field under the four stated assumptions. **Plain language:** once permission to rely is withdrawn, it must not come back quietly; a field anyone can rewrite cannot promise that, because setting it back silently restores a revoked permission and nothing downstream can tell. **What it means:** revocation needs memory, and restoring reliance is still possible — through a new epoch with its own identity, which is the point. **What it does not mean:** improvement is not forbidden; it is routed through recomputation, with the narrowing history retained. The everyday instance is the approval that outlives its authority: signed Tuesday, reorganized Wednesday, still green Thursday. Epoch semantics make Thursday's green impossible without Wednesday's recomputation.

The four defects of section 3.1 now have formal addresses. Staleness is the applicability and currentness gap Theorem 4's epochs govern. Concealed uncertainty is fiber-sharing — Theorems 1 and 2 — where one label absorbs states the estate must tell apart. Self-authorship is Theorem 3's hypothesis in the wild: the write

path unconstrained by truth. Persistence after correction is revocation without memory — the re-expansion Theorem 4 forbids. The architecture in the next section is not a menu of features against these defects; it is the three escapes, composed.

3.5 The three escapes

The architecture developed in the next section combines three structural escapes the formal results leave open: **(1)** a richer retained multidimensional account, so distinctions are stored rather than collapsed; **(2)** a non-settable, evidence-derived verdict, so the record cannot be written independently of the facts; **(3)** epoch-scoped, subset-only reliance with revocation memory, so what was licensed can narrow and be withdrawn with notice, and history is preserved as history rather than as current authority.

Table 4. The formal core at a glance. Each result is a statement about representation under the elected model — never a product effectiveness claim.

Result	Statement	It means	It does not mean
Theorem 1	4 values force ≥ 3 -state sharing; ≥ 9 states in shared fibers	sharing is forced by counting	which merges are harmful (that is C1/C2)
Theorem 2	safety under {C1,C2} forces $ S \geq \lceil G / A \rceil$; = 6 on the full product	six is the floor, paid for by discarding admission	that six values are an adequate design
Theorem 2'	adding C3 forces singleton fibers; $ S \geq 12$	categorical preservation needs the full product or a multidimensional account	twelve values in a UI; display projections stay harmless
Enumeration	4^{12} mappings; 0 avoid both classes	implemented predicates match stated ones	an independent proof of the theorem
Theorem 3	unconstrained truth-independent writes reach $G \times S$	the write path is the deeper defect; vocabulary cannot cure it	a ban on constrained, evidence-derived writes
Theorem 4	$R(e,k+1) \subseteq R(e,k)$ within an epoch; unenforceable by a sole arbitrarily rewritable licensing field (four stated assumptions)	reliance must narrow within an epoch, with revocation memory; the reset counterexample defeats a sole rewritable field, while $2^4 = 16$ is the subset-construction cardinality of the safe, evidence-derived route	a prohibition on improvement across recomputation; a claim about vocabulary size

3.6 What this contributes to computer science

The formal core travels with five bounded contributions to computer science; none depends on artificial intelligence. These five characterize the article's contributions; the architecture, mechanisms, evidence, economics, applications, and commercial consequences instantiate, operationalize, evaluate, or follow from the underlying formal, architectural, and measurement work from which those contributions are stated.

Table 5. Five contributions to computer science, each with its boundary.

Contribution	What it adds	Boundary
Conflation as a property of representation, not practice	A status field smaller than the bound necessarily merges permission-vs-effect-status or effect-status-vs-license; unconstrained settability destroys guaranteed ground-truth information at any vocabulary size. The claimed contribution: the choice of axes, and settability — not vocabulary size — as the deeper defect.	A statement about state representation under the elected axes — bounded and checkable; prevalence across deployed software is not measured here.

Contribution	What it adds	Boundary
A composition proof-obligation framework for decisions	Explicit operators — sequence, parallel, contingent, fuse, retry, compensate, supersede, retract, loop — each carrying proof obligations: safety, liveness, identity, scope monotonicity, provenance, failure propagation, cancellation, compensation, temporal validity, equivalence to a reference semantics.	The contribution is the obligation set, not any operator implementation; composition becomes checkable rather than emergent.
Observation assurance is not consensus	A quorum proves enrolled observers responded, not that the set could detect the event; not-observed becomes did-not-occur only where a calibrated observation topology licenses the inference.	Distinct from Byzantine agreement; prevalence across deployed architectures is not established here.
An assurance measurement framework	Load-bearing predicates — independent, complete, current, sufficient, material — must be measurable or policy-defined; independence is demonstrated as a seven-dimension vector on which common-mode agreement establishes nothing.	Does not prove every such property is fully observable; the reviewed corpus is not enumerated.
Explicit proof, observability, decidability, and trust boundaries	States where proof ends and trust begins — who verifies the verifier, and where that regress terminates by declaration — and governs what may be claimed given undecidability.	Governs around undecidability and unobservability; does not remove them.

3.7 Two bounded design precedents: promises and structured objects

Mark Burgess's Promise Theory provides an established account of policy and cooperation among autonomous components: agents make voluntary promises, and observers assess whether those promises are kept [8], [9], [10]. That work supplies a useful model of commitment where components cannot simply compel one another. Finality Assurance should be read alongside it, but it computes a different object. The object here is current, evidence-derived, party-scoped reliance on a result after occurrence, under an anchored obligation roster, with currentness, withdrawal, and recomputation. A promise or an observation may contribute evidence to that computation; neither is itself the Finality Verdict.

Jeffrey Snover's work on PowerShell provides a second, bounded precedent for preserving structure across a composition boundary [11]. Traditional command shells commonly pass text, requiring each downstream command to reconstruct type and structure; PowerShell instead carries structured objects through the pipeline. Finality Assurance applies an analogous design move at the governance layer. It retains authority, task identity, evidence, occurrence, claim scope, currentness, and reliance as typed governance state rather than collapsing them into a writable status label. The comparison is deliberately limited: Finality Assurance does not reuse the PowerShell pipeline, and PowerShell does not compute the current, party-scoped reliance object defined here. The shared principle is that structure needed by downstream decisions should travel with the object rather than be reconstructed repeatedly from a flattened projection.

3.8 First principles and what follows from them

What a first principle is. A foundational truth, requirement, or irreducible constraint from which a system's structure can be derived without relying on inherited implementation choices — a statement you cannot get underneath without changing what the problem is. Four tests qualify: **fundamental** (removing it changes the problem, not the design), **independent** (not a consequence of another statement), **defensible** (supported by logic, mathematics, a constraint, a requirement, or an observed fact), **generative** (other requirements derive from it). A principle is engineering rather than philosophy only if it continues down a chain able to fail:

first principle → invariant → requirement → mechanism → negative test → retained evidence

Table 6. The twelve first principles. Identifiers are stable across the FAS family; they are not renumbered when classification changes.

ID	First principle	Short explanation, and ground
P1	Reality is independent of its representation.	A status, report, or projection cannot create the fact it describes; an authorized performative act is an issuance event, not evidence another fact held. Ground: ontological — the record can be wrong about the world.
P3	Meaning and identity precede evaluation.	Subject, evidence, policy, authority, scope, and question are fixed before evaluation, or versioned so a change opens a new one. Ground: logical.
P4	Authority, capability, and admissibility are different.	A credential or technical ability never proves the exact transition is permitted now for the stated purpose. Ground: categorial — three facts, none implying another.
P5	Runtime decision dimensions remain separate.	Admission, execution, occurrence, outcome, canonicity, assertion, reliance, applicability: distinct typed decisions. The theorem covers the admission/effect-status/reliance core; the other five rest on the runtime taxonomy — weaker support, stated as such. Ground: informational, and prior to the conflation result — the theorem adds that collapsing the core cannot be repaired downstream.
P8	False, unknown, conflicted, stale, unavailable, and not-applicable are distinct.	Absence of proof is not proof of falsity; each state carries a typed consequence. Ground: informational — merging any two is unrecoverable.
P9	Completeness is relative to an anchored evidence and source universe.	Valid only against a named, derived, signed universe; callers cannot shrink it. Ground: logical.
P11	Historical validity is not current applicability.	Receipt bytes stay immutable; present reliance recomputes against current evidence, authority, policy, time. Ground: temporal.
P13	The subject cannot be its own sole judge.	A producer may emit evidence but cannot alone establish its own correctness or independence. Ground: independence.
P15	Reliance is query-relative, consequence-aware, and scoped.	A finality decision answers one observer-action-purpose-consequence question. Ground: definitional.
P17	Control presence is not control capability.	Load-bearing only after discrimination, nondegeneracy, and currentness are established; claimed interventions must be shown to occur. Ground: evidential.
P19	Local finality does not compose automatically.	Component verdicts compose only under explicit interface, temporal, authority, dependency, and common-cause contracts. Ground: compositional.
P20	Timing, load, failure, false refusal, and liveness are part of correctness.	Safety without usable resolution, timely invalidation, bounded stale reliance, or safe degradation is operationally incomplete. Ground: scope — too late, or refusing everything, is not an answer.

What a further governing principle is. A constitutionally binding proposition downstream of the first principles, stated separately with its ground published, so a reader who accepts the first twelve and rejects a consequence has a specific place to stand. Whatever its ground, it binds as strongly as what it comes from. The ground column carries four non-interchangeable statuses: **entailed**; **declared axiom** (adopted, not proven); **design election**; **implementation construction**. Only entailed rows are derived principles in the strict sense; the others bind by declaration or election. P7b and P18b are declared axioms sharing an identifier with an entailed half — both bind; only one is derived.

```

FIRST_PRINCIPLE != DERIVED_PRINCIPLE
DERIVED != OPTIONAL
DECLARED_AXIOM != PROVED_RESULT
DESIGN_ELECTION != LOGICAL_NECESSITY
P5_PRECEDES_THE_CONFLATION_RESULT

```

Table 7. The twelve further governing principles: the ground of each, stated by type.

ID	Further governing principle	Ground, and its type
P2	One consequential concept resolves to one canonical semantic authority for the evaluation at hand.	P1, P3 — entailed; multiple sources are admissible when governed resolution yields one interpretation per query and epoch.

ID	Further governing principle	Ground, and its type
P6	The transition is the universal governance object.	P5, P15 — design election argued from fit; binding by declaration.
P7	Evidence is multidimensional (P7a) and noncompensatory (P7b).	P7a: P8, entailed. P7b: declared axiom — non-compensation is forbidden by declaration, for stated reasons, not proven.
P10	Domain-model adequacy is continuously governed.	P9, P11 — entailed, with continuous governance the elected mechanism.
P12	Corrections preserve history, conflict, and causal lineage.	P1, P11 — entailed; erasing the predecessor destroys the evidence the correction was needed.
P14	Narrative, claims, and projections from the reporting path cannot originate authority.	P13 — entailed; authorized issuance events are authority acts, untouched by this rule.
P16	Reliance creates a transitive consequence graph.	P15 — entailed; edges compose for consequence, permission does not transfer along them.
P18	Confidence and optimization are advisory, never dispositive (P18a derived, P18b declared).	P18a: P4, P13, entailed. P18b: declared axiom — no score is dispositive; admissibility resolves to typed evidence.
P21	Production claims require exact implementation refinement and outside-in rederivation.	P13, P17 — entailed, with outside-in rederivation the elected construction.
P22	Product surfaces must be non-bypassable for the transitions they govern.	P17, P19 — entailed; one ungoverned path breaks every governed composition argument.
P23	Semantic versions and migrations are explicit.	P3, P11 — entailed; the version of the meaning is part of the identity of the claim.
P24	The canonical output is a typed Finality Account; the Finality Verdict is its action-specific categorical projection.	P5, P8 — entailed, with the typed packaging a design election.

P5 precedes the conflation result — the theorem quantifies over those distinctions and adds that collapsing them cannot be repaired downstream. Publishing the derivations makes the set a constitution rather than a manifesto: reject a derivation and the principle above it is in question; reject a first principle and everything descending from it is. The table below ties each contribution to its constitutional ground, architectural mechanism, evidence surface, and selected adjacent established work — comparisons as boundaries, not claims of supersession; the cited entries are adjacent background, no proposition in this article rests on them, and the monograph carries the full realization map.

Table 8. The scientific synthesis: each contribution's constitutional ground, architectural realization, evidence surface, and selected adjacent established work it should be read against.

Contribution	Ground	Architectural realization	Evidence / test surface	Selected adjacent established work, and the distinction
Representation-level conflation and settability	P1, P3, P5	typed Finality Account; non-settable, query-relative Verdict	Theorems 1–3 and the enumeration; 7-vs-0 acceptance campaign	authorization re-evaluated during use (UCON) [15]; the object computed here is party- and purpose-scoped current reliance, not access — and settability, not vocabulary size, is the proved defect
Decision-composition proof obligations	P19, P20	dependency graph; explicit operators, each carrying preservation obligations	hazard and composition campaigns	program algebra with equational laws (KAT) [16]; this is an obligation framework, not a completed algebra, and says so
Observation assurance is not consensus	P13, P17	contradiction-capable occurrence path; independence as a measured vector	hazard and observer evidence; paired mutation controls	event ordering [12], Byzantine agreement [13], diagnosability of discrete-event systems [17]; the missing predicate is observability of the event, prior to agreement

Contribution	Ground	Architectural realization	Evidence / test surface	Selected adjacent established work, and the distinction
Assurance measurement	P7, P9, P17, P18	required-vs-achieved evidence; class ceilings; seven-dimension independence	evidence batteries and mutation controls	assurance and risk vocabularies (NIST AI RMF) [14]; provenance relations (PROV-O) [18]; here the load-bearing predicates must be measurable or policy-defined, and common-mode agreement establishes none of them
Explicit proof, observability, decidability, and trust boundaries	P13, P21, P22	outside-in rederivation; Four-Corners; bounded claim licenses	deterministic-regeneration campaign and conformance gates	proof-carrying code and the trusted-base question [19]; the boundary — who verifies the verifier, and where the regress terminates by declaration — is itself machine-checkable state

The adjacency claims above are bounded by a dated search protocol executed on 16 August 2026: thirteen recorded queries derived from the seven elements of the composition, run against a general web index surfacing scholarly-indexed material, arXiv, the ACM Digital Library, IEEE Xplore, Springer, MDPI, and the IETF Datatracker, with stated inclusion and exclusion rules; every query, hit disposition, and retrieval date is recorded in the retained search log. The two closest works found — post-hoc execution-compliance certification with a binary conjunctive verdict [21], and pre-action authorization with ordered assurance levels and non-ratcheting policy [22] — each satisfy some elements of the composition and not the full set; the monograph carries the protocol and the element-by-element comparison. The claim remains scoped to the corpus reviewed, not the field as a whole.

Table 9. The five contributions: type, maturity, and the observation that would narrow or falsify each.

Contribution	Type and current maturity	What would narrow or falsify it
Representation-level conflation and settability	Formal result; proved under the elected model (Theorems 1–3 and the enumeration)	A counterexample: a settable status representation satisfying the stated axioms that preserves the specified distinctions or enforces within-epoch narrowing
Decision-composition proof obligations	Framework; architecturally specified — the obligation set, not any operator implementation	An admissible composition producing a higher verdict than its weakest mandatory element under the stated rules
Observation assurance distinct from consensus	Framework; architectural — distinct from Byzantine agreement	A demonstration that quorum agreement alone establishes detection coverage on the measured event class
Measurable assurance predicates	Measurement construct; locally observed on named, producer-graded estates	A field campaign where computed acceptance fails to prevent the failure class it targets at materially higher rates than reported
Explicit proof, observability, decidability, and trust boundaries	Methodology; states where proof ends and trust begins	A claim shown to travel beyond its stated boundary, or a declared boundary shown to conceal rather than govern an undecidable or unobservable step

4. The architecture: computed reliance

The proposal is to stop storing the answer and compute it. A decision declares what it requires — an anchored roster of obligations. Evidence is gathered against that roster and typed by what it can establish. A verdict is computed, never written. Claims and reliance are licensed separately. And the whole computation is redone when anything it rests on changes. This section explains the composition; the monograph carries the complete object model.

4.1 The obligation roster, and required versus achieved

The roster is **anchored** (bound to one governed decision and one applicability universe), **versioned**, **non-empty**, **non-caller-shrinkable**, and **declared by an authority separate from the work**. Anchored completeness is load-bearing: all checks passed must never mean the caller selected the checks; a caller-narrowed roster is refused, not passed. Two pairs must never be conflated:

REQUIRED_ASSURANCE ≠ ACHIEVED_EVIDENCE
REQUIRED_INDEPENDENCE ≠ ACHIEVED_INDEPENDENCE

What a decision requires is declared policy; what the evidence achieves is measured fact; an obligation is discharged only where achieved meets required on both proof grade and independence, dimension by dimension.

The vacuity failure is worth naming because every local estate in which it was tested encountered a form of it: **all checks passed** where the caller supplied the checks. A caller-shrinkable roster converts acceptance into a self-assessment with extra steps — the subset launderer selects obligations it can meet, passes them, and inherits the vocabulary of compliance. Anchored completeness refuses this structurally: the roster belongs to the decision, not the caller; a narrowed roster yields a refused verdict, not a smaller triumph. The roster has a lifecycle of its own, and its governance is half its value: declared by an authority separate from the work; versioned so a decision made under roster v3 remains legible after v5 ships; anchored to an applicability universe so completeness is computed against a denominator that cannot shrink quietly. Obligations enter and leave through governance, never through the caller. In regulated domains the roster is where written policy becomes executable — the policy names the obligation, the roster carries it, the evidence classes discharge it, the verdict enforces it — a chain auditable end to end.

Obligations and their meanings must themselves be machine-evaluable — Semantic Computation™'s office: claims, subjects, authorities, evidence, scopes, and licenses as typed, identity-bound, versioned records rather than prose. No ontology language is prescribed: RDF and typed protocol buffers are both conformant; free text alone is not, however good the text — the requirement is identity-bound, machine-evaluable semantics, not a syntax. Prose remains the human rendering, not the computational substrate. One guard closes the composition: no surface may grade its own work. An adapter producing evidence about outputs its own author generated is an attestation surface in an evidence producer's uniform — separated or ceiling-capped accordingly. Applied to people the rule is unremarkable; the architecture refuses to forget it when the author is a model, agent, or vendor integration.

What any deployment may claim is a function of where its controls sit, and the architecture is deliberately placement-independent: the decision layer integrates at application, control-plane, model-and-orchestration, or infrastructure boundaries, per work class, with no single central runtime. Placement changes the claim,

not the grammar — occurrence evidence from inside the executor's own trust domain supports governed process, not independent occurrence, and the account records which. The strongest available claim — non-bypassability for the governed transitions — is engineering rather than declaration: it requires closing the alternate paths, and a control that can be routed around is present without being capable. The composition also states its own limits. Even fully integrated, the architecture would not establish metaphysical truth, complete world knowledge, solutions to undecidable problems, absence of all unknown adverse evidence, or any improvement in intrinsic model intelligence. It would establish that the computation, decision, action, claim, and reliance process was governed to an exactly declared profile and remained valid at reliance — a narrower claim than the industry usually makes, and the correct one. Undecidability and unobservability are governed around, not removed; not-observed is never converted into did-not-occur.

People are inside this composition by design, with different verbs: humans set policy, hold exception authority, supply domain judgment, resolve contested decisions, escalate — they authorize. What no human does is mutate acceptance state by hand: verification runs as a separate role provably unable to mutate state; producer-verifier disagreement fails closed; an exception is a governed object — held, typed, explained, the missing obligation named — rather than an override. When the evidence is not there and the work is genuinely done, the verdict is held, typed, and explained, converting the missing obligation from an argument into a finding. One governed unit of work ties the figure together. A change is admitted: authority checked, roster anchored, applicability declared. GMRS selects a route — perhaps a local model plus a deterministic verifier, perhaps escalation to a person — under the work's consequence tier and budget. Execution produces typed evidence as a by-product; the account assembles; the verdict computes; a claim license bounds what may be said, a reliance license names who may act; release happens because the verdict permits it. Weeks later a dependency fails: the affected set is computed, the decision recomputes to DEGRADED, its license narrows, holders are notified, and the historical record of the original acceptance stands untouched — every step queryable afterward, none requiring a human to remember anything.

Degradation is where governed estates differ most from brittle ones. When evidence ages, an observer fails, or a provider degrades, the composition forces no choice between ignoring the problem and stopping the estate: verdicts recompute to DEGRADED with the failing obligation named, reliance narrows to the exact decisions affected, and work whose roster is undamaged proceeds. Selective degradation, targeted quarantine, and computed recovery are the operational face of Continuous Finality™ — resilience as a property of the grammar, not an eighth value channel.

4.2 The Finality Account and the Finality Verdict

The **Finality Account** is the retained multidimensional case file — escape (1): per required obligation an epistemic state, a lifecycle state, a proof grade, a proof root, temporal bounds, and query context; nothing averaged, nothing discarded at write time. The **Finality Verdict** is its query-relative categorical projection — escape (2): computed for one exact reliance question, set by no actor. A detached token — FINAL, PASS, CURRENT — carries no meaning apart from the query it answers.

Query-relativity in one example: the same Finality Account, asked may this analyst cite the result in an internal memo, may answer VERIFIED — asked may this release manager route customer traffic on it, BLOCKED, because the second roster requires occurrence evidence and an authority the first never needed. Neither answer is the status of the work; each answers one reliance question — which is why FINAL detached from its query is a category error, not an abbreviation.

4.3 Conservative composition

BLOCKED < UNKNOWN < DEGRADED < VERIFIED
Verdict = meet over the anchored required roster

The verdict is the **meet** — the greatest lower bound — over the required roster: one blocked obligation blocks the result regardless of strength elsewhere; an obligation that does not apply never enters. **Bad news wins**. Four consequences are deliberate: adding a required obligation can never improve the aggregate; improving evidence for an existing one can; strength on one axis may not compensate weakness on another — a declared constitutional choice, not a theorem; and ceilings hold — no volume of lower-class evidence aggregates upward into a stronger class or into authority.

The ceiling rule has a plain reading: a thousand log lines do not become an attestation, a thousand attestations do not become an independent observation, and no volume of any evidence class becomes authority. Each class carries the most it may claim, declared in advance; composition may weaken an answer, never launder one upward — the quiet rule that makes bad news wins safe to automate.

Independence is measured, not organizational. A quorum proves enough enrolled observers responded, not that the set could detect the event — three observers watching the same blind spot agree perfectly and know nothing. Independence is relative to an enumerated failure-model: administrative separation without material separation — a shared foundation model, provider, source estate, prompt lineage, parser, verifier library, benchmark author, or control plane — is not independence; required independence is declared per obligation, exactly as proof grade is.

4.4 Occurrence, claims, and reliance

Occurrence must be established through a path capable of contradicting the executor's own account: a successful command is not an occurred effect, and an agent's report of what it did is evidence of what it claims, never proof of what happened. **Claim licensing and reliance licensing stay separate**: a claim license bounds assertion; a reliance license names the party, purpose, and window — the object that narrows, expires, and is withdrawn.

4.5 Currentness and Continuous Finality™

A verdict is valid at reliance, not forever. **Continuous Finality™** — escape (3) — makes currentness mechanical: dependencies are recorded when reliance is taken, so invalidation is a computation rather than archaeology; when a basis fails, the affected set recomputes, licenses narrow or withdraw with notice, and the historical decision is preserved as a true record of what was known then. A record may exist, may be current, may be currently authoritative — and the third does not follow from the first two. A worked invalidation: a verdict rests on an upstream approval, a calibration, a policy version; the dependency fails. The affected set is computed from recorded reliance edges rather than reconstructed from memory; every decision inside it recomputes; licenses narrow or withdraw with notice; the historical record stays true as history. The downstream estate learns before acting — nobody had to remember who was relying on what, because the reliance edges are that memory.

4.6 Sufficiency and maturity, kept apart from the verdict

The **Four-Corners Test™** asks whether a public conclusion can be reconstructed from the evidence its four corners cite — a sufficiency test, not the verdict. The **Confidence Ladder™** types evidence maturity —

specified, exercised, executed, adversarially tested, independently re-derived or validated, field validated — and its rungs are claim ceilings: each informs scrutiny and permitted claim scope; none sets finality or grants permission to rely. Most results here sit at executed or adversarially tested; nothing sits at independently validated or field validated, and the article says so wherever it matters.

One governed object makes the composition concrete: the feature flag. Ungoverned, a flag is a mutable runtime boolean anyone with write access can flip — a settable status with production consequences. Governed, a flip is an admitted change under an authority, inside an effect envelope, its required evidence declared in advance; it triggers affected-set recomputation, every decision relying on the prior behavior is identified and recomputed rather than silently orphaned, and the flip is an epoch event with a record. A flag flip is a release — and the product in section 8 governs it as one.

4.7 The governed executors: GMRS, constrained learning, MCE

GMRS selects how work is done across routes — models, tools, deterministic computation, humans — under hard constraints from authority, evidence requirements, consequence tier, budget, latency, and the executor's measured failure profile. Admissibility precedes optimization: with no admissible route, the correct output is a governed hold naming the failing condition. A route is not a model name; it is the evidence plan, verification path, stopping rules, and escalation posture attached to the work. The hold is a feature, not a failure mode. A route may be inadmissible because the consequence tier requires evidence no available executor can produce, the budget cannot carry the required verification, or the only capable executor carries a failure profile the work cannot tolerate. The correct output is a typed refusal naming the failing condition — actionable — rather than a silent best-effort by an inadequate route, which is how ungoverned estates convert capability gaps into incidents. Escalation to a person is a first-class route with its own evidence contract.

Routes are contracts, not preferences: the identical model under a different assurance trajectory — different effort, evidence, verification, permissions, stopping behavior — is a different route with a different cost-to-disposition. MCE-established controls enter as constraints: a route whose executor exhibits a failure family the work cannot tolerate is not down-weighted but inadmissible; neither router nor learned policy may waive such a control to improve its objective.

Constrained Policy Reinforcement Learning lets the routing policy adapt within the admitted action space while structurally unable to redefine its own admissibility: it may rank admitted routes, and may not define admissibility, alter evidence requirements, weaken consequence obligations, broaden claim licenses, or promote itself — $\text{LearningAuthority} \cap \text{AdmissionAuthority} = \emptyset$. Promotion of a learned policy is a governed decision with its own evidence; the mechanism must be able to be told no.

MCE — the Mechanistic Conflation Engine — discovers conflation and failure families in agentic work, mechanizes them as candidate detection predicates, and promotes validated detectors into persistent machine-enforced controls under governance, scanning continuously thereafter. It originates no authority, cannot promote its own findings, and cannot establish that unknown open-world failure modes are absent; no controlled benchmark of the scanner on external workloads is reported — its evidence status is internal operational use, stated in section 8.

What MCE scans is the surface the formal core names: execution conflations (successful execution read as governed occurrence), method conflations (a failed test read as the intended control reached), authority conflations (a record's presence read as issuance; a valid signature read as an authorized signer), trust-domain conflations (separate process read as separate trust domain), custody conflations (a secret removed

from current state read as removed from history) — each a non-equivalence, mechanized as a detection predicate and, once validated, enforced continuously rather than remembered.

Table 10. The composition at a glance: each capability's technical role, and why anyone cares.

Capability	Technical role	Consequence
Finality Assurance™	computes current reliance from typed evidence over an anchored roster	unsupported work cannot acquire authority by assertion
Semantic Computation™	makes obligations and their meanings machine-evaluable	policy, scope, and done become executable rather than interpretive
Continuous Finality™	recomputes when any basis changes	historical validity cannot silently remain current authority
GMRS	selects the admissible execution route	premium capability is reserved for work that demonstrates a need for it
Constrained Policy RL	ranks admitted routes; cannot admit or promote itself	potential compounding economics without self-relaxing safety
MCE	converts discovered failure families into persistent controls	known failures need not be rediscovered session by session
Work-management product	produces governed evidence and state as work executes	replaces status archaeology and fragmented authority

4.8 The composition in one figure

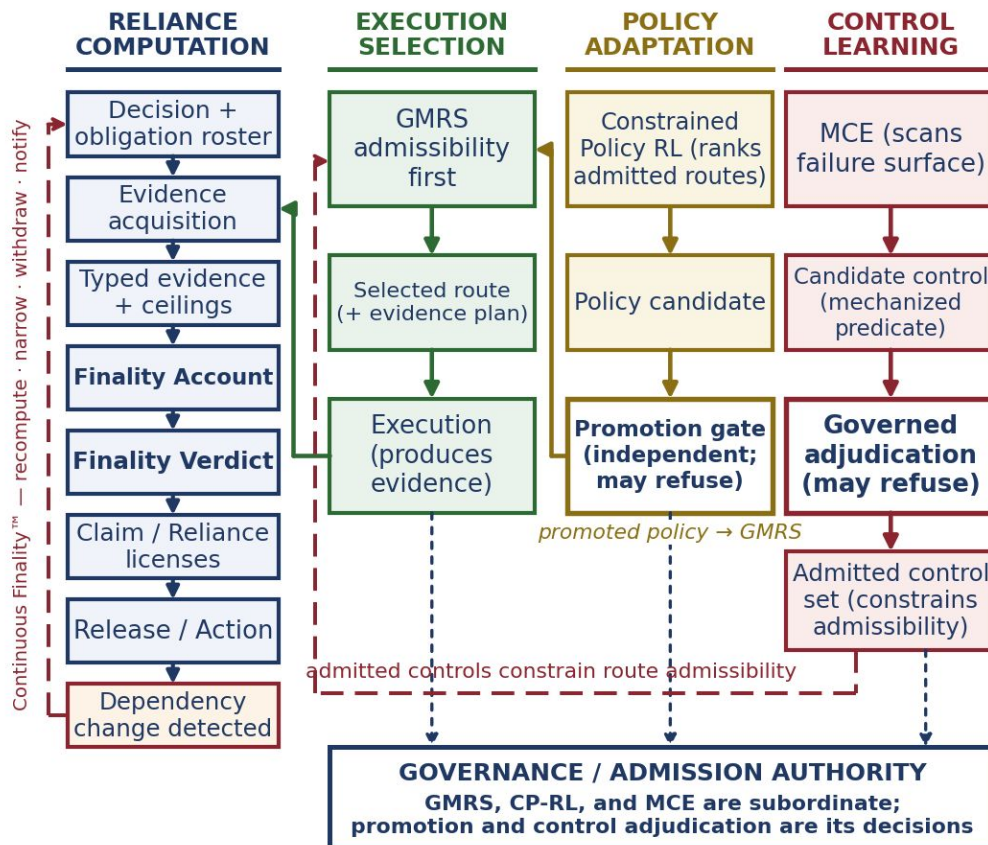


Figure 2. Computed reliance in four lanes. The reliance computation runs left; execution selection, policy adaptation, and control learning run beside it and remain subordinate to governance/admission — CP-RL reaches GMRS only through an independent promotion gate, MCE reaches enforcement only through governed adjudication, and neither writes evidence, the Account, or the Verdict. Reserved implementation detail is not depicted.

5. Benchmark evidence

Architecture is not utility. This section asks whether separating these objects and computing acceptance actually changes observable outcomes. Every number is a property of the exact campaign that produced it; all campaigns are producer-graded on named local estates; material adverse results are carried with the claims they qualify, at full strength. Findings are presented in a deliberate hierarchy — decision quality first, the current adverse live envelope prominent, learning last because it is open. **No third party has reproduced anything.**

How the campaigns relate: the 2,520-answer acceptance campaign and the 30-task legitimate-work battery measure the decision layer — what ships and what is refused. The recorded 36-task battery and the live 240-task campaign measure routing — how work is executed and at what cost. The hazard, mutation, reproduction, and safety-gate campaigns measure the assurance machinery itself. They are separate estates with separate denominators, and no number from one may be quoted against another's baseline; the hierarchy below is prominence, not chronology.

Table 11. The evidence map: which campaign answers which question, and its status. All producer-graded; none independently validated; none field validated.

Campaign / battery	Question it answers	Status
2,520-answer acceptance campaign (six models, paired)	does computed acceptance change what ships?	observed; governed arm 0 false accepts, ungoverned arm 7; bounded to recomputable tasks
30-task legitimate-work battery (two conditions)	does evidence, not correctness, gate admission?	observed; the 90/90 refusal cost is the designed price
36-task recorded battery (frozen)	can cheaper exact routes compose valid dispositions?	observed; saturated — cannot resolve quality
240-task live cloud campaign (paired)	do the economics survive live conditions?	observed; ADVERSE on quality and safety; descriptive
Governed adaptation	what incremental value does governed adaptation add over the governed static baseline while preserving decision quality and evidence discipline?	bounded internal and benchmark support; incremental longitudinal effect not isolated; external generalization open
Hazard / mutation / reproduction / safety-gate campaigns	does the assurance machinery itself hold?	observed; causal attribution only where paired

Producer grading is disclosed on every campaign for the same reason the adverse result is prominent: the article's claims must survive its own evidence discipline. Grading rules were fixed before outcomes were seen, refusals are typed, and every attempt produces a verdict record — a refusal is an issued verdict, and a missing verdict is a pipeline defect, so the denominator cannot be quietly narrowed by unexamined work. What producer grading cannot supply is independence, and no phrasing in this article converts the one into the other: the grading is honest, and it is the author's.

5.1 Decision quality

Across **2,520 graded answers** from **six real language models** on verifiable tasks, paired task-for-task, the ungoverned accept-on-claim arm shipped **seven wrong or fabricated answers** as completed work; the governed arm, computing acceptance from executed evidence, shipped **none**. One local model returned $47 + 47 = 74$, confidently, in four of five attempts — and the ungoverned arm accepted it. All seven arose on two of the six models; the other four produced no false accept in either arm — the observed benefit was concentrated on the two models that produced the failures on this estate.

A second campaign measures the failure that ungoverned pipelines miss rather than the one they commit. On a realistic multi-surface change, a smaller frontier model omitted a required schema contract and reported completion in five of five attempts — pooled omission rate 0.80 across four prompt phrasings. Governed finality, computed from executed evidence rather than from the agent's narrative of what it did, shipped that broken work zero times of five and drove every change to convergence with a root-cause fix. The scope is one task family, paraphrase-tested: omission, not error, is the failure class demonstrated.

Boundary, stated exactly. The tasks were chosen because a verdict about them can be recomputed from executed evidence; the zero is a **soundness property of recomputation** on this task class — not a general accuracy claim, not a claim about work whose correctness cannot be recomputed, not a field result. Producer-graded on a named local estate, like everything in this section.

5.2 Evidence-qualified admission

On a separate 30-task legitimate-work battery run in two conditions, the governed arm refused **0 of 270** answers that were correct and evidenced — no false refusal was observed among the 270 correct, evidenced responses on this battery — and the campaign does not measure the process cost of producing and evaluating that evidence. It also refused **90 of 90** answers that were correct but carried no evidence. That second number is the point, not an embarrassment: it is **designed evidence discipline**, the price of an acceptance decision computed from evidence rather than from the answer's plausibility. A correct outcome is not evidenced acceptance; the 90/90 row is what that distinction costs, published with the benefit rather than beneath it.

The two conditions define the discipline's operating characteristics on this estate: zero friction where evidence accompanies correct work, total friction where it does not — by construction. An operator should read 90/90 as the pipeline's demand for evidence plumbing, not lost work: a governed pipeline declines work a human would have waved through — the verdict is computed from what was established, not what is probably true.

What would move each result up the ladder is concrete: the acceptance result needs an independent estate — someone else's tasks, someone else's grading; the admission result needs adversarial evidence-forgery attempts, not just presence/absence; the routing results need unsaturated batteries with power calculations. Each step is designed, none claimed in advance; the numbers stay exactly where their evidence puts them until the next campaign moves them. Negative and null findings publish under the same rule as adverse ones. The safety gate failed its declared utility criterion — it held a substantial share of work while showing no observed false acceptance over an unpublished trial count, a direction rather than an elimination, holding cost reported with it. No isolated adaptive-learning result is reported anywhere in this article or its source; bounded internal and benchmark evidence supports the governed adaptation architecture, the incremental longitudinal contribution of adaptation itself is not yet isolated, and implementation-specific promotion history and update-control evidence are reserved. A program reporting only its safety improvements is not reporting its results.

If a reader carries three numbers out of this section, they travel together: 7 against 0, 90 of 90, 21 against 1 — computed acceptance changes what ships; the discipline has a real price, published; the current live routing envelope loses quality and safety to the frontier comparator, the article's own instrumentation saying so. That is the honest state of the evidence: a mechanism that works where measured, a cost declared, a boundary named and targeted. Every number can be audited to its origin: the monograph [1] carries each campaign's estate description, arm definitions, denominators, grading rules, refusal taxonomy, and artifact

digests; the deterministic-regeneration campaign establishes that the retained artifacts regenerate the declared outputs. A reader disputing a figure has a named place to stand — the property the architecture demands of the work it governs, applied to the article's own evidence.

Two design choices recur and deserve their reasons. **Pairing**: wherever both arms can face the same task, they do — task-for-task on the acceptance and live campaigns — removing task-mix as a confound; every difference is a difference on identical work. **Counts over ratios for rare events**: 21 against 1 is reported as counts because a ratio built on one comparator event is unstable — the article quotes the counts and lets the reader divide, rather than quoting a 21× that one more comparator event would halve. The same conservatism runs the other way: the governed arm's zero false accepts is never quoted as a rate of zero — no observed event is not zero probability; it is 0 observed over the declared denominator, and the article says the denominator every time. What a false accept is, operationally, keeps the stakes concrete. The seven ungoverned false accepts were answers shipped as completed work that were wrong or fabricated — including $47 + 47 = 74$ delivered confidently in four of five attempts by one local model. The governed arm's advantage was not knowing better; acceptance was computed from executed evidence the answer had to survive. All seven arose on two of the six models, and four produced no false accept in either arm — the benefit concentrated where the executors actually failed on this estate, the pattern a computed failure profile exists to capture: concentration, not proportionality.

Table 12. What would move each result, and in which direction the next evidence step runs.

Result	What would strengthen it	What would falsify or bound it further
Decision quality (7 vs 0)	an independent estate and grader; unsaturated harder batteries	false accepts under computed acceptance on recomputable tasks
Evidence admission (0/270; 90/90)	adversarial evidence forgery attempts refused	evidenced-incorrect accepted, or evidenced-correct refused at scale
Recorded composition (24 vs 36)	unsaturated batteries with power calculations	substitutions failing off the frozen battery
Live economics (75 refusals at <half cost)	powered replication; observed-cost accounting	savings migrating to different-outcome pairs
Adverse envelope (21 vs 1; 0.883 vs 0.988)	calibrated selective routing closing the gap under power	the gap persisting under calibration — which would bound the routing thesis itself
Governed adaptation (bounded evidence; increment not isolated)	an observed, justified admission under the same gates	admissions that degrade quality or evidence discipline

5.3 Recorded route composition

On a **36-task frozen battery**, governed routing matched the strongest static comparator using **24 rather than 36 model calls** — **twelve exact deterministic substitutions**, where a computation replaced a model invocation outright — and **no frontier calls** in the governed arm. Both compared arms recorded 1.000 correct disposition. **Boundary**. The battery **saturated**: both arms at ceiling, it cannot resolve a quality difference, and it licenses no parity claim — saturation is not parity. What it establishes is route composition: some valid dispositions are reachable through cheaper, exact routes on this recorded battery. The twelve substitutions deserve their name — exact deterministic substitutions, the governed arm answering with a computation whose correctness is a property of arithmetic, not sampling; not a cheaper model imitating an expensive one. Such substitution moves cost without touching the quality question — exactly why the saturated battery, unable to resolve quality, can still establish composition.

One further recorded property is licensed by the source, with its axes: on the declared comparison, governed routing was the only arm of eleven not dominated across cost, accepted work, disposition, safety, and call composition — non-dominance over that recorded battery's arms, not superiority on any axis.

Table 13. Recorded 36-task battery (frozen, saturated). Modeled dollars under a declared list-price model.

Measure	Governed routing	Strongest static comparator
Correct disposition	1.000	1.000
Model calls	24 (12 exact deterministic substitutions)	36
Frontier calls	0	36 — the comparator is the frontier arm
What the battery can establish	route composition on this recorded battery	the comparison ceiling

5.4 Live outcome-equivalent economics

The cost basis deserves a paragraph of honesty rather than a footnote. Modeled dollars in both routing campaigns are computed under a declared list-price model — public per-token prices applied to counted tokens — the basis a reader can recompute. Observed usage cost on the live campaign ran roughly sixty-two times lower on the governed arm and seventy-two times lower on the frontier arm than the modeled figures; the retained record does not apportion that divergence among pricing, caching, batching, credits, or other account effects, so the modeled row is comparable only within its own table — it ranks the arms under one declared rule and prices nothing in anyone's budget. Ratios survive the caveat better than absolute dollars; the article quotes ratios.

Table 14. Paired 240-task live cloud campaign. The adverse rows are the finding; modeled dollars are comparable only within this table.

Live campaign metric	Governed route	Static frontier comparator
Tasks	240	240 (same battery, paired)
Correct disposition	0.883	0.988
False accepts	21 (rate 0.0875)	1 (rate 0.0042)
Model calls / tokens	199 / 122,165	240 / 152,075
Modeled cost (declared list-price model)	\$0.9491	\$1.8871
Identical correct refusals (both arms)	75 — at less than half the comparator's modeled cost	75

5.5 The adverse live result: the current operating envelope

The same live campaign is why no end-to-end superiority claim appears in this article:

21 false accepts under the governed route vs 1 under the frontier comparator
0.883 correct disposition vs 0.988

End-to-end superiority is not established by this live campaign. Substituting away from the frontier comparator reduced calls, tokens, and modeled cost — and reduced accepted-outcome quality and safety on this battery. Correct disposition here counts accepted-correct, escalated-and-accepted, and safe-decline outcomes; an arm's rejection of its own incorrect answer counts as neither correct disposition nor false accept (five such events for the governed arm, none for the comparator; counted as correct dispositions the rates would read 0.904 against 0.988). The reading is direct: this is exactly the trade-off the routing layer exists to expose, surfaced by its own instrumentation. A layer that could not surface it would be worse, not better. The defensible conclusion is selective, calibrated routing under measured constraints — not that cheaper models substitute for frontier models. This envelope is the concrete target of the powered follow-on

study. Three readings of the adverse result are wrong and worth refusing explicitly. It does not establish that governed routing is inferior in general — one campaign, one policy, one estate, descriptive and single-run. It does not establish that the architecture failed — the architecture is what surfaced the numbers, task-paired, with the arms' costs beside them. And it does not license the mirror-image overclaim that better calibration makes superiority follow — precisely what the powered study exists to test. What it establishes is an envelope: on this battery, this policy substituted away from premium capability where premium capability was buying quality and safety.

5.6 Governed adaptation: what is established and what remains open

Constrained Policy Reinforcement Learning is not presented here as an isolated learning-performance result; it is an architectural component of the governed system whose bounded benchmark, prototype, and internal-operational evidence is reported below. Its public invariant is that adaptation operates only inside an independently determined admissible action space and cannot author its own admissibility, evidence requirements, or promotion authority.

Established. The governed adaptation layer has been exercised as part of the product family's benchmark, prototype, and sustained internal operating environment. Those estates provide bounded evidence for governed routing, operational control, decision quality, evidence discipline, cost-to-disposition effects, and the failure-to-control feedback loop. They are evidence that the architecture produces useful behavior under the stated conditions.

Open. The public evidence does not presently isolate how much incremental benefit is attributable specifically to adaptation rather than to the surrounding governed-routing and control architecture, nor establish the durability of that incremental contribution over longer horizons or independent external estates. Those are the longitudinal questions the powered benchmark and external-validation program are intended to answer.

Implementation-specific promotion history, candidate counts, branch coverage, gate-level outcomes, estimator outputs, and update-control evidence are reserved.

Assurance-machinery detail, briefly. The hazard campaign planted six scenario designs across 720 trials — 360 hazard, 360 matched clean — blocking all 360 hazards and wrongly refusing none of the clean work; four designs share one window-opening mechanism, so the count is domain coverage, not six independent mechanisms. The release-regression registry's 46/46 detection is one measurement stated twice — zero false accepts is its complement. The author-side regeneration result — 53/53 and 109/109 declared outputs regenerated, archive rebuilds reaching byte-identical fixed points — establishes the evidence estate is reconstructable from what was retained, by the author; author-side regeneration, not independent validation, and never counted as such.

5.7 Supporting assurance

Table 15. Supporting assurance evidence. Producer-graded; author-side regeneration is not independent validation.

Campaign	Result	Boundary
Hazard containment	360/360 planted hazards blocked over 720 trials, with a matched clean arm showing no good work wrongly refused	4 of 6 scenario designs share one window-opening mechanism; scenario count is domain coverage, not mechanism diversity
Release-regression mutants	46/46 detected; zero false accepts is the detection count restated in complement	a release regression registry carried across releases, distinct from the campaigns above

Campaign	Result	Boundary
Paired mutation discrimination	8 matched pairs: clean control passing, mutant refused, expected delta observed — over 2 shared clean-control executions (one serving seven ordinary mutants, one the authorized-epoch case)	causal attribution follows for these 8 pairs alone, not automatically for unpaired mutants
Deterministic regeneration	53/53 and 109/109 declared outputs regenerated by the author; archive rebuilds reached byte-identical fixed points	author-side deterministic regeneration; no independent external validation exists
Safety gate	under a three-times-fault simulation the gate showed no observed false acceptance over an unpublished trial count — a direction, not an elimination — while holding a substantial share of work	unpublished denominators; the gate failed its declared utility criterion, reported as such

6. The economics of Accepted Work

If accepted work can be separated from generated work, model calls, tokens, and activity cease to be sufficient economic units; the question becomes what it costs to obtain work the organization can actually count.

6.1 Cost-to-disposition

Cost-to-disposition is everything required to reach any disposition: execution, model and token costs, retries, verification, escalation, review, rework, audit reconstruction, failed attempts, governance overhead. A failed or blocked run costs money and produces zero accepted-work value — everything stays in the denominator.

One boundary must be governed for any of this to mean anything: **what counts as one attempt**. An attempt-denominated metric moves by splitting one attempt into three, merging three into one, or reclassifying a failure as a precursor. The attempt boundary is declared by the roster's authority, versioned with it, and not adjustable by the party the ratio describes — so a low token price, high activity volume, or green dashboard cannot flatter the outcome metric.

The interplay between channels is where governed economics differs from cost-cutting. Avoided loss and throughput pull against naive cost reduction — the live campaign's adverse rows are the canonical case: the cheapest route bought its savings with false accepts. Reclaimed expert capacity and latency pull together; delegation option value compounds as decision classes become testable for delegation. Non-additive, because the channels share causes — one mechanism, several surfaces of value. FinOps and token economics are conserved as inputs: cost observability, unit pricing, capacity planning, and consumption discipline feed the denominator. What changes is the numerator's authority — spend per activity gives way to spend per outcome the organization is entitled to count; the one new requirement is that acceptance conditions defining counted work be declared, versioned, and non-gameable, because every downstream economic quantity inherits their integrity.

The channels are also the measurement plan: each maps to quantities a field engagement can baseline and re-measure — the table's field column names them. The seven channels name what value would look like; section 9's engagement protocol measures whether it appears, per decision, against a frozen baseline. The finance owner's summary: keep every FinOps discipline; add one governed numerator; refuse every economic claim arriving without its denominator definition, cost model, and safety columns; and treat any efficiency figure whose quality column is missing as the live campaign's lesson in miniature — on the one battery where cost fell furthest, so did safety, and only the safety columns said so.

6.2 Accepted Work, and its unit

Accepted Work is work actually allowed to count under the declared acceptance conditions — not merely correct, not merely shipped. **Accepted Work per Dollar™** changes the unit of economic analysis from activity to finality-permitted outcome:

$$\text{AwPD} = \frac{\sum (\text{accepted}) w(a)}{\sum (\text{attempted}) C_{\text{disp}}(a)}$$

weighted Accepted Work ÷ full cost-to-disposition (higher is better)

AwPD is a discipline for comparing routes within one declared model — accepted-work definition, cost model, false-accept and false-refusal behavior attached — never a headline number for comparing vendors;

without those attachments an AWpD figure is not interpretable, and this article publishes none — two estates reporting identical AWpD, one counting anything a reviewer initialed, the other only roster-discharged typed evidence, share arithmetic but not meaning. The live campaign is the standing caution: on the very battery where the governed route's cost fell, its quality and safety fell too — lower cost is not superiority, and AWpD read without the safety columns would have flattered the wrong arm.

6.3 Why FinOps inputs are insufficient

Tokens, model calls, GPU hours, model tier, and seats are input costs — essential as inputs to the full cost of acceptance. None is the economic output, and optimizing them without an acceptance-derived numerator rewards exactly the failure the adverse result documents: cheap activity that does not survive the acceptance decision.

Table 16. The operating-metric replacement: activity inputs remain essential as inputs; outcome metrics acquire the authority.

Input / activity metric	Outcome metric it feeds
model calls, tokens, GPU time	Accepted Work per call and per dollar
tickets closed; builds green	correct dispositions; release-authorized work
review count	Accepted Work per expert hour
model utilization	opportunity capture at held quality
audit effort	reconstruction cost per relied-upon decision

6.4 Seven channels of enterprise value

The enterprise value of computed reliance takes seven forms; this article measures only some of them, and the table carries each channel's meaning beside what was and was not measured. The channels overlap and **cannot be summed** — an impact model, not a measured equation — and operational resilience is cross-cutting rather than an eighth additive pool. Institutionalized proof spend — United States public-company external audit fees were approximately \$21.7 billion in 2024 [4] — indicates the mechanism's scale without being a claim on that spend.

Table 17. The seven channels: measured here versus field-measured. Channels overlap; rows may not be summed.

Channel	Measured in this article?	What a field engagement would measure
Avoided loss — unsupported work cannot create rework, harm, unauthorized action, or remedy costs	mixed bounded decision-quality evidence (7-vs-0 favorable; 21-vs-1 adverse); no loss amount measured	incident, rework, and remedy cost against baseline
Reclaimed expert capacity — experts stop reconstructing routine proof, reserved for judgment and contested cases	no	expert minutes per disposition
Lower cost-to-disposition — the same correct decision via exact computation, a cheaper route, or a correct refusal	recorded and live campaigns, modeled costs	full loaded cost per accepted disposition
Latency / opportunity value — a defensible decision inside its window	no	time to accepted work inside the decision window
Accepted-Work throughput — more safely countable work per unit of money, time, attention	no	accepted units per period at held quality
Delegation option value — work refused for automation becomes testable for delegation	no	work newly delegable under computed acceptance
Trust / revenue premium — proof over assertion lowers friction with customers, auditors, insurers, regulators	no	audit effort, dispute rates, counterparty friction

7. Where the same structure appears

The mechanism transfers wherever consequential work needs authorization, evidence, acceptance, currentness, and defensible reliance under a different domain contract — what counts as occurrence, which authority applies, what burden of proof holds, and what degradation is permitted all differ. These are applicability analyses and field hypotheses. **No result is reported in the domains below, and no safety-critical certification result exists.**

Two rows of the table repay a slower read. In healthcare, the conflation is institutional: the portal's approved, the payer's authorized, the chart's indicated, and the pharmacy's dispensable are four different assertions held in four systems, and today a person is the join. A computed admission decision does not practice medicine — it assembles the roster (indication, interaction, dose, authorization, license), types the evidence each element already has, and computes what may proceed, holding what cannot, with the lapsed element named. In trading, the window is the point: a defensible decision after the window is a record, not a trade. Computing the reliance question moves assurance from after-the-fact reconstruction into the window — and when the basis fails mid-window, withdrawal with notice is the difference between stopping an action inside the decision window and letting an unsupported action proceed, with whatever operational, contractual, or legal consequences follow.

The public-benefits row carries the volume lesson: determinations are high-count, rule-dense, and audited in arrears, which is precisely the profile where improper-payment pressure accumulates — eligibility evidence, documentation, and authority each checkable, none jointly computed at determination time. The software row carries the self-application lesson: it is the one domain in this article where the operating model has been exercised end to end, which is why its evidence is deepest and why the article resists letting that depth redefine the product's category. One domain supplies the worked machinery; six supply the shape of the problem; none of the six carries a field result yet.

One horizontal pattern is worth working end to end because it is common across the classes considered: audit reconstruction. Today an audit is an archaeology project — sampling records, re-deriving what was relied upon, interviewing the people who remember. Under computed reliance the artifact the auditor needs is the Finality Account itself: what was required, what discharged it, who relied, under which roster version, with which evidence — retained at decision time, not reconstructed after. Evidence reconstruction becomes a query over the retained decision record, permitting broader and more systematic coverage; independent audit procedures, tests of completeness, and controls over the Finality Account itself remain necessary. No field engagement has yet measured that conversion; the mechanism that would be measured is fully specified, which is exactly the boundary between architecture and outcome this article keeps.

One consequence of computed reliance reframes the stakes across every domain above: a standard-of-care effect. Wherever a computed, evidence-derived acceptance decision exists and is obtainable, the post-incident question could shift from is assertion acceptable to why was the obtainable precaution not taken — the same ratchet that made reconciled ledgers, change control, and independent audit ordinary. This article makes no legal claim and reports no instrument requiring machine-verifiable acceptance evidence today; the observation is conditional, not prescriptive: once a precaution becomes technically obtainable, its availability can become part of post-incident operational, contractual, regulatory, or legal scrutiny. An organization evaluating this architecture is therefore pricing two futures, not one — the cost of adopting computed acceptance, and the cost of explaining its absence.

Three theses about how the ecosystem develops are stated in the monograph and argued against before any is adopted; the compressed form belongs here. That providers will absorb this — argued against because provider assurance and destination reliance are different questions, and a provider grading its own work is the attestation surface the grammar refuses. That the market will standardize on observability instead — argued against because observation without declared obligations produces dashboards, and dashboards are the telephone game with better graphics. That regulation will force the outcome regardless — argued against as a plan because regulation names duties, not mechanisms, and the estate that waits for the duty to name the mechanism will build it under a deadline. None of the three is dismissed; each is a live path whose evidence would change this article's applicability analysis, and the monograph states what would change each.

Table 18. Six flagship domains: the same decision structure under different domain contracts. Conditional consequences, not demonstrated transfers.

Domain	The decision	What can become conflated	Evidence needed — and what must remain current	What changes if reliance is computed
Healthcare	may this order proceed for this patient	approval status vs clinical evidence vs payer authorization	indication, interaction, dose, authorization, license checks — must remain current: authorization; formulary; clinician license	routine authorization evidence assembled mechanically, exceptions routed for review; reliance withdrawn when a basis lapses
Finance / trading	may this trade execute inside its window	booked vs confirmed vs settled; approved-then vs valid-now	mandate, limits, confirmations, market-window evidence — must remain current: the mandate; the limits; the window itself	a defensible decision inside the window; withdrawal with notice when the basis fails mid-window
Aviation / mission	does this authorization still hold	issued-then vs valid-now; effect vs license	authority chain, environment state, currentness evidence — must remain current: the authorization itself, continuously	stale authority is computed as stale, before action
Software / cybersecurity	may this build be released; is this closure real	green pipeline vs releasable; closed finding vs discharged obligation	tests, scans, provenance, rollout evidence, receipts — must remain current: dependency state; threat surface; runtime posture	the writable acceptance surface is retired; evidence decides
Manufacturing / industrial	may this batch be released	batch record vs sensor calibration vs quality evidence	calibrated measurement chains, batch genealogy — must remain current: instrument calibration; process window	a lapsed sensor blocks the affected release or narrows permitted operation per the declared degradation policy; unaffected work continues
Public benefits / payments	is this determination payable	workflow status vs entitlement evidence	eligibility evidence, documentation, authority — must remain current: policy version; superseding determinations	improper-payment pressure met with computed admissibility

7.1 Frontier models

This architecture does not replace frontier models, and the live adverse result is the proof that the boundary matters: cheap, local, and deterministic routes are not universally adequate, and on the live battery the frontier comparator's quality was the standard the governed route failed to match. GMRS's office is the opposite of model replacement — it can reserve premium capability for work that demonstrates a need for it, substitute exact computation where computation is the correct executor, and hold work for which no admissible route exists. Provider-side assurance and destination-party reliance are different questions: what

a provider establishes about its own runtime does not answer what a relying party may establish about a specific result — and full integration establishes governed process, not truth.

Integration is tiered, not binary. At the shallowest tier a provider's output is simply typed evidence with a declared ceiling; deeper tiers admit provider-attested execution context, then governed invocation with evidence capture at the boundary, then full in-path governance. Each tier is a different claim about what a relying party can establish — and none of them, including the deepest, converts a provider's self-assessment into occurrence. The model is deliberately provider-neutral: the same tiers describe a frontier API, an open-weight model on local hardware, or a deterministic tool.

A bounded illustration. The provider/destination distinction stopped being hypothetical in July 2026, when Hugging Face (16 July) and OpenAI (21 July) each published a primary disclosure of the same event: an autonomous evaluation trajectory that left its intended boundary, persisted across a long horizon, and reached production infrastructure [5], [6], with a later retrospective technical timeline [20]. The structural reading both disclosures support is the one this article is about: the run's own account of itself was not the instrument that caught it, and nothing in the acceptance path was positioned to contradict the executor. It is a class argument, not a criticism of either organization — both disclosed publicly, and they are named only because a reader cannot check the analysis without knowing whose record it is; the monograph works the incident through the framework at failure-class level.

7.2 Across the ecosystem

The same separation touches every layer without naming winners. Open and local models gain a route to consequential work they cannot reach on self-attestation. Frontier providers gain a destination-side acceptance layer that makes their quality legible at the point of reliance. Hyperscalers and platforms gain an acceptance-derived unit — Accepted Work — beside consumption metrics. Agent frameworks gain an external answer to a question they cannot answer from inside: whether an agent's claim about its own work may be relied upon. Observability, GRC, provenance, and domain systems follow the product's own disposition grammar rather than a permanent-overlay story: where retained, they supply typed evidence or historical record; functions the work-management product performs natively may instead be subsumed or replaced; and no retained system remains an independent authority over acceptance or finality. Domain systems of record keep history; a computed answer about present reliance is added beside it.

Beneath the six flagships, the same few horizontal patterns recur across these consequential-work classes: reconciliation work that exists only because no record is canonical; exception routing that exists because admissibility was never computed; audit reconstruction that exists because evidence was not captured as work executed; currentness watchdogs that exist because nothing recomputes; and delegation reviews that exist because no one can say what a machine would have to prove. Each is a compensation for a missing computed answer, and each is priced in expert time. That is why the applicability claim is horizontal while every efficacy claim in this article stays bounded to its named campaign.

8. The governed work-management product

The architecture becomes an operating model in a governed work-management product, defined deliberately broader than software: **a governed operating system for consequential human, AI-agent, machine, software, transactional, and physical work**, whose primary unit is persistent organizational work state — not a ticket, a pipeline run, or a routed task. It governs what is authorized; which requirements apply; which dependencies are satisfied; how work executes; what occurred; what the evidence establishes; who may accept; what is released; whether prior acceptance remains current; what is superseded; how remedy and reliance unwind propagate; and how organizational capability changes as a consequence. **Software engineering is a deep embodiment, not the product boundary.**

8.1 Prescriptive at the reliance grammar, not at the domain

The product is prescriptive and opinionated about reliability semantics, not the business. It prescribes the meaning of **authorized, ready, evidenced, accepted, released, closed, final, superseded, and current** — consequential work is not free to invent incompatible definitions per department or tool. Customers control application and business logic, architecture, infrastructure, domain tests and risk controls, policies, specialized tools, model and provider choice, and deployment topology — including operation inside their own control boundary. **No second grammar:** no status field, dashboard, ticket, integration, or agent may become a parallel authority over finality; a new capability must prove it collapses into the existing model before any new origin of meaning. The refusal list is the clearest description of the product — prescriptive means refusing things. The design refuses to let anyone: set finality, readiness, or acceptance by hand, from any interface, by any actor; mutate canonical state outside the registered boundary; start consequential execution from an incoherent plan; treat a passing pipeline, green dashboard, scanner run, vendor claim, or agent summary as an acceptance decision — evidence, never a verdict; let a claim travel further than its license, or publish one stronger than the evidence licenses; move a verdict with a confidence score; let a note, memory entry, or agent trace become authoritative by existing; or add a parallel grammar without proving reuse first. The stated reason is arithmetic as much as principle: if N independent generators each state something stale with probability f , at least one is wrong with probability $1 - (1 - f)^N$; a single canonical origin returns that to its own staleness rate — an illustration showing direction, not magnitude — it reuses the independence assumption the opening flags as failing in real estates.

Table 19. Where variation belongs: the customer controls the domain; the operating model prescribes reliance semantics.

Customer-controlled	Prescribed by the operating model
application and business logic; architecture	evidence and acceptance semantics
domain-specific tests and risk controls	test-first and prompt-test proof requirements
deployment topology; infrastructure	claim and finality grammar; authority separation
specialized tools; model and provider choice	the meaning of ready, accepted, released, closed, final
policies and their content	required evidence classes; the prohibition on second grammars

8.2 What happens to existing tools

The disposition model is capability-based, four-way, and operator-controlled: **REPLACED** outright, where the product natively performs the function and the incumbent is no longer technically necessary — authority replacement (settable status surfaces, readiness trackers, reconciliation, evidence binders) and native functional replacement alike, the operator free to retain an incumbent for compatibility, preference, migration, policy, or redundancy; **SUBSUMED**, its semantics and authority moving into the governed model;

RETAINED as subordinate executors, scanners, evidence producers, stores, or systems of record — significant work, no independent authority over the meaning of done; **MIRRORED** during transition as explicitly non-authoritative projections. Functionally replaceable does not mean mandatorily removed; retention by choice does not create architectural dependence.

8.3 The deepest embodiment: AI-first software delivery

Software engineering exercises the operating model hardest — agent-produced change is nondeterministic, expensive, and side-effecting. The embodiment ties each capability to the reliance grammar, not a feature list: governed work state with computed lifecycle; test-first and prompt-testing disciplines; generated, impact-selected tests with full-suite fallback; CI admission, merge, and capacity governance; release, deployment, and runtime assurance; evidence capture as a native product of execution; and application-factory inheritance — generated systems arrive inside the organization's governed delivery model — testing baseline, admission semantics, evidence contracts, release controls, finality model — from their first commit, each retaining its own runtime truth. These are design commitments of the operating model, stated as such; the controlled results in section 5 measure the decision layer, not this surrounding capability set. **The assurance baseline the factory inherits** is broader than test-first alone, every class an evidence producer feeding admission and finality rather than a green light: unit and contract testing; property-based testing; fuzzing; end-to-end simulation with fault injection; differential oracles; continuous regression; source, dependency, container, infrastructure-as-code, secrets, and software-bill-of-materials analysis; agent and agentic-workflow security testing; adversarial review — impact-bounded selection deciding which checks a change requires, full-suite fallback when impact is uncertain; a generated application inherits the baseline as infrastructure, its output evidence for its own acceptance decisions, not a status.

The scope of governed work is broad and knowingly bounded. Work is anything whose outcome someone will rely on — a change, decision, document, release, incident, experiment, model promotion. Not everything is governed: a question, draft, or exploration carries no reliance, needs no roster — forcing admission machinery onto it would be the approval theater this architecture exists to remove. The moment an output will be relied on, acted on, shipped, or learned from, it crosses into governed work — selective governance is designed; governing everything fails as much as governing nothing. Above work items, the organizing object is the capability: a durable ability with an owner, evidence obligations, a dependency graph, and a computed build, stabilize, keep, or retire disposition; the graph computes which capabilities block which releases and what must mature first. One boundary is easy to get wrong: capability acceptance is a rollout over work acceptance, never a second proof engine — a capability model that starts deciding whether work is acceptable has grown the second grammar the whole design exists to prevent.

Adoption requires no big-bang replacement, and the destination does not preserve the legacy estate by default. The first placement touches no application code: existing tools' outputs become typed evidence for a decision computed beside them — value arrives there, each further step optional and independently useful. From there the operator elects the boundary: where the product natively performs a function the incumbent may be replaced outright; where a specialized system adds value it remains a subordinate executor, evidence source, or projection surface; boards and dashboards may persist as explicitly non-authoritative mirrors, retired when their coordination value lapses — no retained or mirrored surface keeps standing as an authority over the meaning of done. No central runtime, no bottleneck team owning the shared model: the decision layer deploys per work class, team, tenant, or region, and the no-second-grammar rule keeps federation from becoming fragmentation — many deployments, one grammar, zero duplicate origins of meaning.

Why the bespoke-CI model changes under AI-first development. Traditional software organizations treated CI/CD as a local snowflake: each estate accumulated its own pipeline grammar, status semantics, approval conventions, and integrations — tolerable while human engineers produced relatively deterministic change at human speed. Agent-generated work changes the risk calculus: a system that can produce code, prose, attestations, status changes, and side effects at machine speed cannot safely let each pipeline invent a parallel grammar for what ready, accepted, released, or final means. Domain logic remains local; the reliance grammar does not. The product therefore makes test-first and prompt-test proof, evidence production, merge queuing, multi-node runner fan-out, capacity-aware dynamic CI/CD shaping, release, deployment, and runtime assurance native governed capabilities — and where those capabilities fully cover an incumbent function, that tooling may be replaced outright, per the disposition model above.

What the operating model computes for the organization is memory with authority. Every attempt produces a verdict record; a refusal is an issued verdict; a missing verdict is a pipeline defect; recorded reliance edges make blast-radius a computation before a risky change, not a war-room exercise after it, and supersession preserves lineage — what replaced this, why, and who relied on the predecessor stay answerable. Because capability dispositions, evidence obligations, and accepted-work economics live in one governed graph, the questions executives ask — what can we safely delegate, what blocks the release, what did accepted work cost — become queries against state already held. Why test-first is a reliance property rather than a methodology preference: a test written after an implementation is evidence the implementation can pass a test; a test whose red state is re-executable — re-run against an implementation-absent state and observed to fail — is evidence the obligation preceded the work. The product treats the second as the provable form of the-test-came-first, for code and governed prompt behavior alike. The same logic runs down the delivery chain: merge queues make landing collision-safe while validation stays parallel; capacity-aware shaping makes validation depth a governed decision, not a timeout artifact; deployment profiles and staged rollout because repository and operational finality are distinct facts; runtime assurance because acceptance does not end at a green build — each capability the reliance grammar applied to one more surface where a settable status used to live. The deployment boundary is part of the product's definition. The operating model runs inside the customer's chosen control boundary — local, customer-hosted, containerized, hybrid, multi-region, edge, or air-gapped where supported — surrendering neither execution environment, evidence, policy, nor data domain; the grammar is fixed, and everything it governs remains the customer's.

8.4 Self-application, classified exactly

The product family built the product family: the governed product governed its own construction from early in development, and the failures that work exposed were abstracted into the standards family that now governs it — the product supplied empirical pressure; the standards, normative authority. This establishes **sustained internal operability**, fit to consequential development work, and a loop converting discovered failures into durable controls — **internal operational evidence, the discovery environment; not independent validation, not external production validation**. During sustained internal development use, material conflation and agent-process issues have been identified and cataloged at an average rate of approximately 14 per day — a recurring source of observed failure patterns convertible into durable controls rather than rediscovered session by session. That rate is an observed internal identification rate: producer-graded, including retrospective discovery; not an agent error rate, not a steady-state population rate, not a complete-detection rate; and identification is not adjudication, cure, or closure. The underlying register, denominators, and record census are retained in the product family's internal custody.

One recorded failure-to-control pass (monograph production appendix):


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observed: build and gates green while a declared edit had not occurred
corrected: A_GREEN_BUILD ≠ THE_INTENDED_RENDERED_CHANGE
control: output-level witness on the rendered artifact → standing gate set
recurrence: the same witness refused two further incomplete edits
    
```

Which way authority runs keeps self-application honest: the requirements were discovered in building and operating the product; their normative authority came from surviving abstraction from the exposing implementation — first-principles tests, formal and evidentiary constraints — not from the product having done them. Once admitted, the standards govern the product, not the reverse; conflicting product behavior is a defect, not an amendment — and that admission record is the author's own account, unexamined by any third party.

9. What would settle the open questions?

The powered benchmark. The follow-on study tests what this article's evidence cannot: generalization beyond the named estates; decision quality and false-acceptance/false-refusal behavior under statistical power; route economics off the author's estate; longitudinal adaptation; preservation of safety and evidence constraints under adversarial and held-out scenarios. Its concrete target: the adverse live envelope — 21 against 1, 0.883 against 0.988.

Design commitments follow from this article's own boundaries: power calculations before data collection; estates held out independently of the policy under test; paired designs wherever pairing is honest; false acceptance and false refusal reported as first-class outcomes beside cost, never netted; adaptation evaluated longitudinally with promotion gates observable in both branches; adverse findings carried at full strength with the claims they qualify. Where logged-policy evaluation is appropriate, established off-policy estimators provide one family of methods for estimating candidate-policy value under explicit support assumptions [3]; their use does not alter the requirement for independently governed admission. A study that could only confirm would not settle anything.

One governance property spans both programs: the evidence that would most improve what can be claimed is evidence Expound Laboratories cannot generate for itself — an estate authored by somebody else, a workload nobody here designed, an auditor with no stake in the answer. That is the next rung on the ladder this article uses to type every claim.

The research agenda is not private to Expound Laboratories: a 2026 survey of 257 agentic-validation studies finds temporal validity and runtime evidence maintenance under-developed across the field [7] — precisely the currentness and evidence-maintenance axes this architecture is built on. The powered benchmark and the field program are aimed at questions the field has named and not yet answered: assumption currentness at reliance time, evidence kept live against dependency change, and trajectory-level assurance composed from step-level evidence without inheriting its gaps.

Field validation: one consequential decision. The engagement model is deliberately small. A partner (1) selects one decision that matters; (2) freezes its current baseline; (3) writes its obligation roster — an exercise that tests whether a unified roster already exists; (4) names the evidence classes that discharge it; (5) defines correct disposition; (6) defines false acceptance and false refusal; (7) measures time to accepted work; (8) measures expert minutes per disposition; (9) measures full cost-to-disposition; and (10) retains the evidence under partner control. A successful engagement establishes a field result on the partner's own

decision under the partner's own baseline — evidence no internal campaign can supply. Engagement duration is not reported; the unit of field validation is one consequential decision.

9.1 Safety-critical and mission-critical validation agenda

The general design is necessary but not sufficient for the abstract's domains; before computed reliance may carry weight in clinical, transportation, energy, industrial, or defense contexts, all of the following — no completion of which is reported in this article — remain:

1. domain hazard and failure-mode analysis; common-mode and barrier-independence assessment;
2. integrity- or assurance-level mapping; timing, liveness, and worst-case-latency bounds;
3. deterministic fail-safe behavior; degraded-mode and selective-continuity testing;
4. human-factors and operator-interface validation, including erroneous-refusal and unavailable-evidence handling;
5. recovery, rollback, compensation, and remedy paths; cyber-physical occurrence instrumentation;
6. regulatory and certification engagement in each domain's own regime; independently authored scenario estates; independent external validation by organizations not involved in the technology's development;
7. long-duration monitoring for drift and stale reliance; domain-specific evidence the layer improves outcomes without weakening existing safety cases.

Until each item is discharged for a named domain, no conversion from potential to demonstrated is available to any reader.

Table 20. What would earn a high-stakes claim: domain class by domain class. No efficacy is claimed in any row.

Domain class	Additional proof needed	What would count as success	What remains outside FAS
Clinical / medical	hazard analysis, clinical workflow, human factors, regulatory engagement	reduced false acceptance without harmful refusal or delay	clinical judgment and regulatory authority
Transportation	timing determinism, integrity levels (SIL/DAL), fail-safe and degraded modes, independent barriers	bounded decision and withdrawal behavior under faults	certified control logic
Energy / industrial	common-mode analysis, selective continuity, recovery paths	affected-set isolation without unsafe continuation	existing protection systems
Defense / mission	authority currentness, communications degradation, human command	correct authorization and withdrawal under mission timing	command authority
Cross-domain	independently authored estates, long-duration monitoring, independent external validation	evidence-defined improvement without weakening the existing safety case	the domain safety case itself

10. Conclusion

The chain this article has walked is short. Generation became abundant; defensible reliance stayed scarce. The scarcity is structural. Under the elected formal model, small status projections necessarily collapse specified governance distinctions, and unconstrained settability creates a separate write-path defect; the broader problems of evidence, authority, currentness, and reliance are addressed by the architecture rather than proved by the counting result. Five bounded contributions to computer science carry that chain — representation-level conflation and settability, decision-composition proof obligations, observation assurance distinct from consensus, measurable assurance predicates, and explicit proof and trust boundaries — which characterize the underlying formal, architectural, and measurement work the rest of the article instantiates, operationalizes, or tests. The escape is to compute the answer: a retained multidimensional account, a non-settable evidence-derived verdict, epoch-scoped narrowing reliance. Measured on named estates, computing acceptance changed which work shipped, made evidence a real admission condition, and exposed — rather than hid — a live routing envelope where cheaper execution was not appropriate for the work. The same operating model reads across consequential human, agent, software, machine, transactional, and physical work, and the governed work-management product operationalizes it, with software delivery as its deepest current embodiment. What generalizes is now an empirical question with a designed instrument: the powered benchmark, and field validation one consequential decision at a time.

Spent through the seven channels, the consequence is concrete. Safer: unsupported work stops acquiring authority, and withdrawal reaches relying parties before action. Faster: defensible decisions arrive inside their windows instead of behind their reconstructions. Less expensive: the same correct dispositions reached through exact computation and correct refusal, with reconstruction spend converting to evidence plumbing. More resilient: degradation narrows the affected set instead of stopping the estate. And newly delegable: work refused for automation because nobody could prove the outcome becomes testable for delegation — the frontier the whole argument exists to move. Each claim carries the evidence status its section assigned it; none is a field result yet.

Pick one decision that matters. Write the roster. Whatever happens next, an organization that has written down what a consequential decision actually requires is ahead of one that has not.

Self-application coda

Table 21. Self-application: what internal use establishes, and what it does not.

What internal self-use establishes	What it does not establish
sustained internal operability of the operating model	independent external validation
a recurrence-to-control feedback loop that converts discovered failures into durable controls	independent external validation
applicability to consequential development work	external deployment efficacy
the discovery environment for the standards (which are now normative over the product)	universal superiority

Disclosure statements

Data and code availability. The public layer of this article is intentionally sufficient to evaluate its scientific claims, architectural invariants, and evidence boundaries from the text; internal regeneration bundles and campaign artifacts are retained as custody evidence, not distributed; the independent external validation route is described in section 9. **Funding.** This article was produced by Expound Laboratories; no external funding is stated. **Competing interests.** The author holds the intellectual property and commercial interests described in this article; this disclosure is vendor-authored. **Authorship.** Kenneth L. Cheney is the sole author of record. **Version and errata.** This article is a governed subject; supersessions and corrections are recorded in the release custody chain, and the current subject identifier appears in the provenance section and the subject footer.

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Every reference below supports either a claim carried by this article or an explicitly identified adjacent-work comparison, cited at the point of use; the monograph carries the full 41-entry bibliography.

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Evidence and claim boundary. All benchmark results are properties of the exact named campaigns, batteries, simulators, and profiles described in the monograph; producer-graded on named local estates, not claims of production performance, field efficacy, customer outcome, or generalization; no third party has reproduced them. No patent has issued; no examination has concluded; nothing in this article asserts that any claim is valid, that any third party infringes, or that any party is free to operate. Statements about the public and reserved disclosure boundary are engineering design intent, not conclusions of law. This article notes that reciprocal open-source terms — such as the GNU Affero General Public License, version 3.0 — remain under consideration for GMRS and Constrained Policy Reinforcement Learning; no licensing model is committed, and the article discloses no product readiness, release timing, or development status, and makes no claim of external availability or distribution.

Threats to validity, consolidated. Construct: correctness is producer-graded on tasks chosen because a verdict about them can be recomputed from executed evidence; the measured construct is recomputable acceptance, not field correctness. Internal: every campaign is author-run on named local estates; grading rules were fixed before outcomes were seen, but the grading is the author's and no independent grader exists. External: results are properties of the exact named campaigns, batteries, simulators, and profiles — not claims of production performance, field efficacy, customer outcome, or generalization; no field deployment and no high-stakes domain is claimed. Statistical: denominators are bounded and declared; the recorded 36-task battery is saturated and cannot resolve a quality difference, and the live campaign is a single paired estate without a power calculation. Conflict: this article and its evidence are vendor-authored; the independent estates and validation named in section 9 are the stated remedy, not a completed control.

The public and reserved surfaces, and the licensing paths between them. The disclosure boundary of this article is deliberate and two-layered; the table states it in one glance.

Table 22. Public and evaluable; reserved and accumulated; the commercial paths between them.

Public and evaluable	Reserved and accumulated	Commercial path
problem formulation and taxonomy; the formal model and its bounded results; the twelve first principles and the twelve further governing principles; the runtime-decision separation; abstract FAS, GMRS, CP-RL, and Continuous Finality™ roles; evidence modes and benchmark methods; provider-neutral integration patterns; the research agenda	proprietary implementation methods and machine representations; proprietary decision, composition, routing, calibration, adaptation, and control mechanisms; private semantic models, accumulated failure intelligence, evaluation assets, and anti-bypass methods; deployment-specific integration, security, trust, and operational mechanisms; unreleased source, private data and estates, internal tooling, and operating methods; all other confidential implementation and operational know-how	standards implementation and conformance; a possible release of the routing and constrained-learning layer under reciprocal open-source terms — such terms remain under consideration and are not committed; if released, final terms set at release, no trademark license, and no reserved rights transfer with the code; a commercial embedding license for organizations preferring not to take network-copyleft obligations — availability, scope, terms, timing undisclosed; design-partner validation of one consequential decision

The public layer is intentionally sufficient to evaluate the scientific claims, architectural invariants, evidence boundaries, and commercial relevance of the technology. It is intentionally insufficient to reproduce the proprietary products or reserved implementation. The implementation, accumulated operational intelligence, and other protected material remain reserved. Organizations seeking to deploy, embed, or integrate these capabilities should engage Expound through the licensing, conformance, and design-partner pathways described here. Publication grants no implementation right or license except as expressly provided under a separately executed agreement or applicable released software license.

Where to go deeper. The extended monograph carries the complete first-principles and further-governing-principles tables, the full conflation taxonomy, complete proof derivations, every campaign's genealogy and arm-level records, the application atlas, the economics treatment, the product exposition, the licensing and design-partner programs, and the paper-applied-to-itself production ledger, with public/reserved boundaries stated.

Provenance of this article

This article is a governed derivative — subject FAS_CORE_TECHNICAL_ARTICLE_v20, superseding v19, v18, v17, v16, v15, v14, v13, v12, v11, v9, and FAS_CORE_TECHNICAL_PAPER_40P_v8 — of the monograph release candidate, whose PDF is identified by digest in reference [1]. The frozen derivation-source pair (PDF SHA-256 620afa9574563dc7..., DOCX SHA-256 30dcbc43e1aa437b...) are operator re-saves of that release candidate artifact — DERIVATION_SOURCE ≠ MONOGRAPH_RELEASE_CANDIDATE_ARTIFACT: byte-different, verified content-equivalent probe-by-probe at derivation, equivalence recorded in the companion derivation report, which also carries the derivation protocol, proposition reconciliation, omission accounting, and qualification package, in controlled custody and available for qualified review. This v20 subject is qualified against the monograph release candidate v99 named in reference [1], which carries cures adopted after the derivation source was frozen; discrepancies resolve in favor of the monograph subject named in reference [1].

Appendix — Reliance maturity self-assessment

This appendix is a worksheet rather than an argument, carried from the monograph. Thirteen questions, each scored 0 to 4, covering the dimensions this article argues to matter; the selection is the author's and has not been validated against outcomes. Completion time is not measured here; scored against one decision, it produces two things: a number that says roughly where you stand and, more useful, the specific rows where

you scored lowest — the rows to work on first. Score one consequential decision, not your organization as a whole: averages across decisions hide exactly the variation that matters. Run it again for a second decision: comparing row patterns and floor bands across decisions can be useful; the total score itself is only a within-decision longitudinal progress measure.

Table 23. The scoring scale.

Score	What it means
0 — Absent	The capability does not exist. Nobody could answer the question.
1 — Asserted	Somebody declares it. The declaration is the record, and it is not checked.
2 — Documented	It is written down and reviewable, but assembled after the fact and by the party being examined.
3 — Evidenced	Evidence exists at the moment of change, is typed, and is separable from whoever produced the work.
4 — Computed	The answer is derived from that evidence on demand, cannot be written by an actor, and is withdrawn when its basis fails.

Table 24. Reliance maturity self-assessment: the thirteen questions.

#	Question	What a 4 looks like	Score
1	Can you produce the list of what was required before this decision was made?	A declared obligation roster exists, is versioned, and is authored by an authority separate from the work.	—
2	Is permission to act distinguished from evidence that the act occurred?	Authorization and occurrence are separate records; neither can stand for the other.	—
3	Is occurrence established by something other than the actor's own account?	Occurrence evidence comes through a path capable of contradicting the executor.	—
4	Can the person or system that produced the work influence whether it is accepted?	The producer has no write path to the verdict and no control over the evidence that grades it.	—
5	When a claim is made, is it bounded to a party, a purpose, and a period?	Every claim licenses a named party for a named decision until a named condition changes.	—
6	If a supporting fact became false today, what would happen?	Dependencies are tracked, the verdict is recomputed, and every relying party is notified and unwound.	—
7	Can weak evidence in quantity substitute for strong evidence?	Each evidence class carries a ceiling fixed before composition; no accumulation crosses it.	—
8	Can a confidence score, model output, or self-report move the decision?	Confidence is admitted as an argument and excluded from the verdict.	—
9	How many paths exist by which this decision can be reached?	Every path is enumerated and governed; an ungoverned path is treated as a defect, not an exception.	—
10	What happens when evidence is missing rather than negative?	Absence produces a typed hold naming the missing obligation, never a default to yes or a blanket stop.	—
11	Do independent checks share components with what they check?	Independence is declared per dimension against known common modes, and claims are bounded by what is established.	—
12	Do you know the full cost of reaching a disposition, including refused and reworked attempts?	Cost to disposition and accepted value are both measured; the ratio is a governed operating metric.	—
13	Do the controls you rely on stay current with how the work actually fails?	Failure families are established from what is observed, become controls that run without anyone remembering them, and are re-measured when the model, agent, or tool doing the work changes — noting that no controlled benchmark of the automated scanning layer is reported here.	—

Reading your score. Two numbers come out of the worksheet and they answer different questions, so take them in order. Your **band** is the lowest score among the thirteen rows — not the average and not the total. Your **total** is a progress measure, useful for comparing the same decision against itself six months later and for nothing else. The floor rule is not severity for its own sake: a chain of obligations is only as strong as the weakest one in it, and a single ungoverned path, or one obligation discharged by a self-report, defeats every claim built on the other twelve rows. A rule keyed to the total would say otherwise — nine rows at 4 and four

at 0 sum to 36, and reading that as Documented would promote a decision with four capabilities missing. The total cannot see a zero, which is precisely the thing worth seeing.

Table 25. What your lowest row means.

Lowest row	Band	What this says, and what to do first
0	Ungoverned	At least one capability is absent, and it defeats the claims resting on the rest regardless of what else you scored. First move: find that row and give it any answer at all, even a weak one.
1	Asserted	Reliance rests on people saying so. In the author's experience this is a common starting point, and it is not a criticism — it is what the tools reviewed for the monograph support. First move: answer question 1. Write the obligation roster.
2	Documented	The record exists but is assembled afterwards by the party being examined. Audit is a project rather than a query. First move: push questions 3 and 4 upward — get occurrence evidence that does not depend on the producer's account.
3	Evidenced	Evidence is produced at the moment of change across every row. The remaining gap is that somebody still decides. First move: questions 5, 6, and 10 — bound the claim, track the dependencies, type the holds.
4	Computed and continuous	Every row is derived from evidence, withdrawn when its basis fails, and none of it is written by an actor. If you scored here honestly across two different decisions, we would like to hear from you — the design-partner program is intended to run with a small number of partners at a time, and organizations already operating at this level are the ones it most wants to learn from.

Table 26. The total, read as progress rather than as a level reached.

Total	Illustrative profile at this total	Heuristic interpretation
0–12	Mostly absent or asserted	The total is low because the capabilities are, not because one row drags it down; the floor and the total agree here, and either may be read first.
13–25	Asserted, with documentation emerging	A total in this range with a zero in it is a different situation from one without — the floor governs; this number only tracks movement.
26–38	Documented, with evidence emerging	Progress against a previous score for the same decision is the useful reading; comparison against another organization's total is not.
39–47	Evidenced, with computation emerging	Computation is emerging across rows. A decision in this range may still carry one or two rows well below the rest — the rows the floor exists to surface.
48–52	Computed across nearly every row	Only at a total of 52 do the floor and the total necessarily agree; anywhere below it — including 48, which twelve rows at 4 and one row at 0 also produce — the floor may still be zero, and the floor governs.

One caution about both numbers: a self-assessment carries exactly the weight of one — a conversation starter with your own teams, not a certification, and this article applies the same discipline to its own reported evidence. A high total with one zero is worse than a middling total with no zeros — one ungoverned path or one self-graded obligation defeats the claims built on the rest. Read the lowest row first, the total second.

About Expound Laboratories™



Meaning, made computable. Finality, made governable.

Expound Laboratories™ develops research, standards, and computational systems concerned with Semantic Computation™ and Finality Assurance™. The work is motivated by a distinction that grows more consequential as computational systems become more capable: producing an output is not establishing what it means, and neither is determining whether it has earned a defined status for reliance — three questions that ordinary workflows collapse into one:

What was produced? ≠ What does it establish? ≠ Under what conditions may it be relied upon?

Semantic Computation concerns the explicit representation and computation of meaning-bearing structures — claims, evidence, context, requirements, obligations, dependencies, authority, policy, provenance, contradictions, permitted transitions. Finality Assurance addresses the separate question of when work has satisfied the applicable evidence, policy, authority, scope, and temporal conditions for a governed disposition. The first develops the semantic state; the second governs its disposition.

Finality Verdict issued ≠ Finality satisfied

The mark is the natural exponential, e^x , read two ways. Mathematically, e is Euler's number and the function's rate of change equals its current value — a compact object implying a much richer structure of direct and indirect relationships, the representational analogy intended. Semantically, e stands for Expound and x for the unresolved subject: a question, a claim, a decision, a requirement, a system state. Read as a progression, the mark says: expound the unknown, compute the meaning, govern the finality. An uppercase or stylized E is a brand form of the lowercase mathematical e .

The analogy is representational, not a claim about implementation: nothing in this article computes a scalar or matrix exponential as its governing algorithm, and the mark should not be read as asserting that it does. The exponential also carries the limitation the architecture exists to address: it has no notion of completion, sufficiency, authority, or reliance, and the ability to keep generating does not establish that output is finished.

Expansion ≠ Finality

Scale ≠ Correctness

Confidence ≠ Authority

Expound means to unfold and make meaning explicit; Laboratories adds the requirement to test it — hypotheses, controlled computation, evidence and provenance, measurement, adversarial testing, reproducibility, falsification, refusal. The combination is deliberate in both directions: semantic elaboration without experimental discipline amplifies a wrong premise; a closure mechanism over underspecified semantics authorizes a result whose meaning was never established. Finality here is not permanent, context-free certainty but an earned status inside an explicitly bounded jurisdiction of reliance — why every result in this article is reported with the estate that produced it.

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What may a named party rely upon, about a named result, for a named purpose, now?

Pick one decision that matters. Write the roster. Freeze its baseline; name the evidence classes that would discharge it; define correct disposition, false acceptance, and false refusal; measure time to accepted work and full cost-to-disposition. An organization that does only that much will know more about its own reliance than most audits ever surface.

Three doors. A researcher can challenge this article — the formal results, the enumeration, and the campaign structure are stated to be checked, and an independent estate moves the evidence ladder in whichever direction it points. A design partner can validate one consequential decision under its own baseline. A platform or vendor can pursue standards implementation, the routing and adaptation layer — for which reciprocal open-source terms remain under consideration, without commitment — or a commercial licensing relationship, to the extent an agreed scope includes the relevant reserved capabilities — no availability, scope, terms, or timing are disclosed here.

Whatever happens next, an organization that has written down what a consequential decision actually requires is ahead of one that has not.

The extended technical monograph — Finality Assurance™ for Reliable Decision Quality (241 pages)

complete derivations • campaign genealogy • industry atlas • product exposition • licensing programs

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